

Long–Short Portfolio Return Book

Cumulative return of each factor's long–short portfolio, net of 10 bp transaction cost.
269 factors over 120 months, ordered by in-sample CAGR.

HOW TO READ

Each curve starts at 0 % in July 2015 and compounds monthly long–short returns. The line is coloured by the factor's style; the figure at the right edge is the period-end cumulative return.

FACTOR STYLES

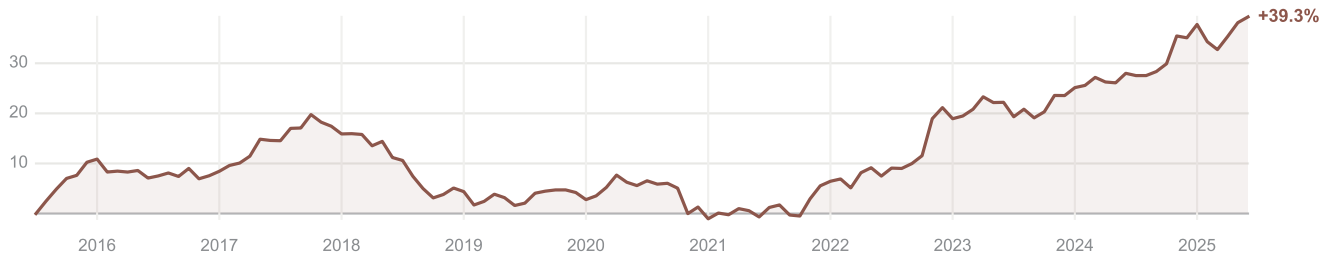
● Valuation	41	● Price Momentum	55
● Earnings Quality	55	● Size	1
● Analyst Expectations	24	● Historical Growth	55
● Capital Efficiency	28	● Volatility	10

001 In current model portfolio (50 factors)

001 ● Num of Qtrs of Pos Chgs in TTM Cash Flows/TTM Sales

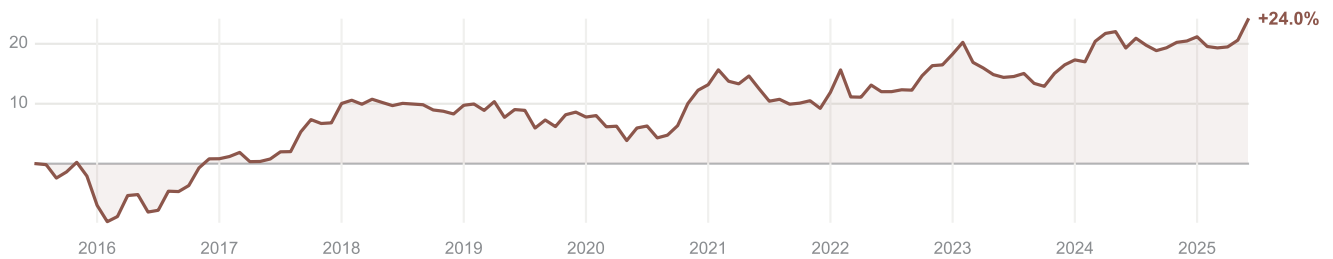
Historical Growth

CAGR +17.2%

**002 ● Trailing 12-Month Receivables and Inventories/Trailing 12-Month Sales**

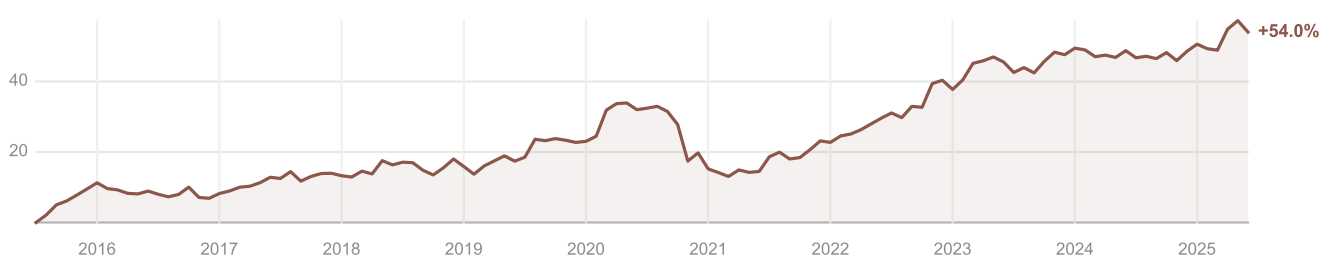
Historical Growth

CAGR +15.4%

**003 ● Number of Consecutive Qtrs of Pos. Changes in TTM Cash Flows**

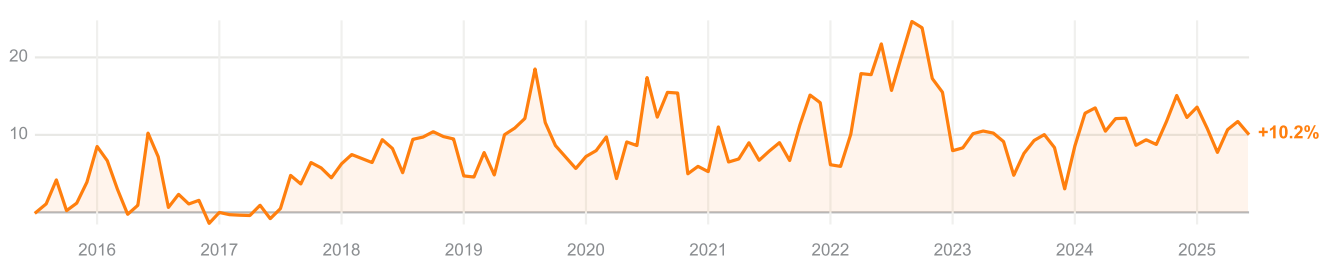
Historical Growth

CAGR +12.3%

**004 ● 50 Day to 200 Day Stock Price**

Price Momentum

CAGR +12.2%

**005 ● Change of Asset Turnover**

Historical Growth

CAGR +11.8%

**006 ● Current Liability to Price**

Valuation

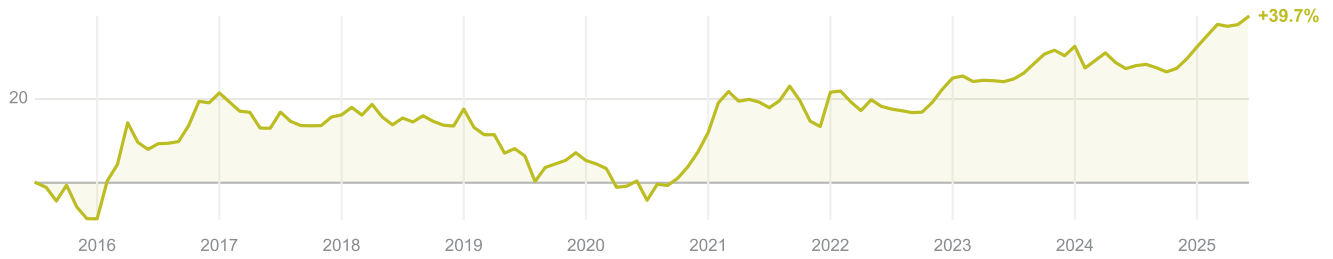
CAGR +11.7%



007 ● **Tobin's Q Ratio**

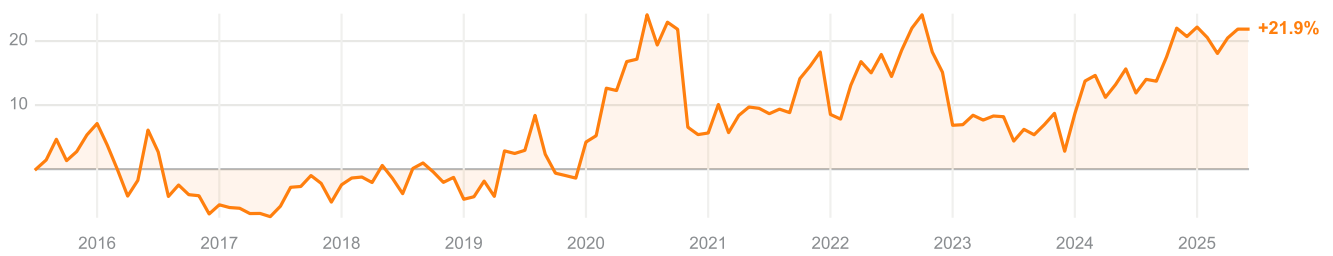
Capital Efficiency

CAGR +11.5%

008 ● **39W Lagged Return**

Price Momentum

CAGR +11.4%

009 ● **Pretax Income Per Share to Price**

Valuation

CAGR +11.3%

010 ● **Operating Earnings to Price Ratio**

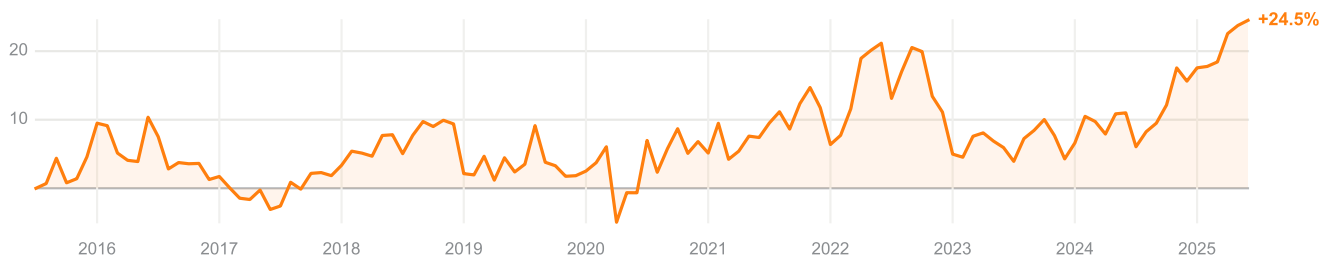
Valuation

CAGR +11.2%

011 ● **26W Relative Price Strength**

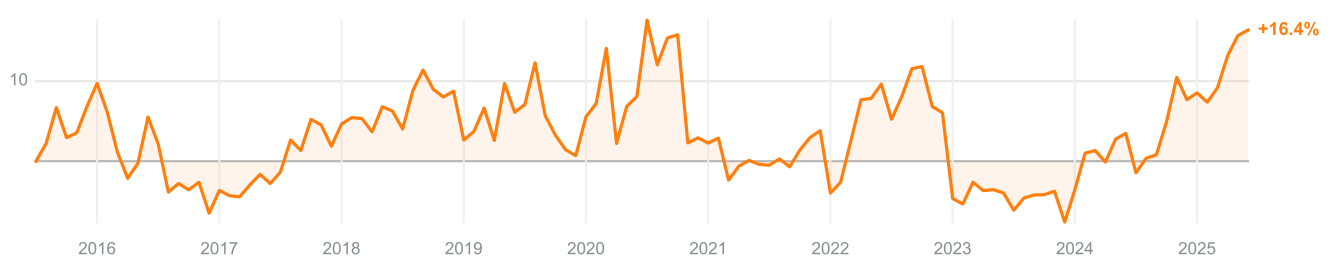
Price Momentum

CAGR +11.1%

012 ● **9 Month Price Momentum**

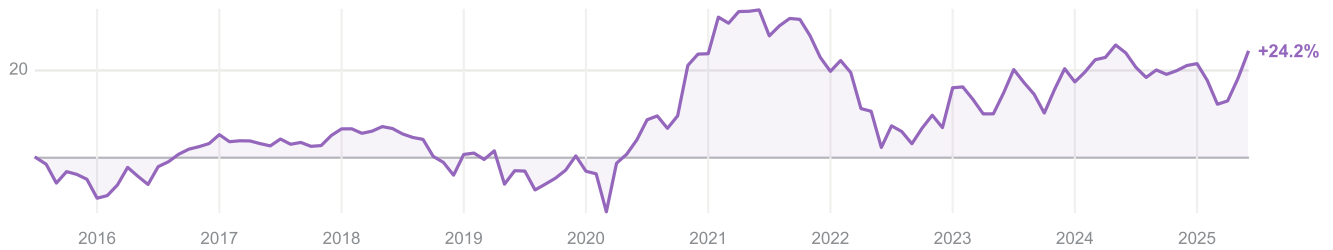
Price Momentum

CAGR +11.1%



013 ● **12M Realized Price Volatility**

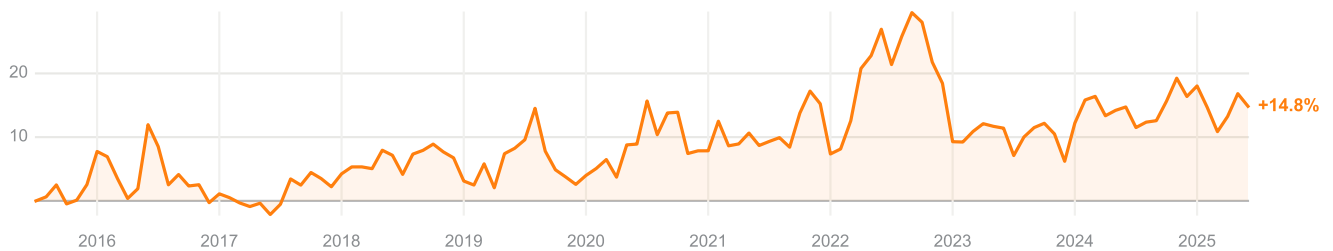
Volatility CAGR +11.0%

014 ● **Slope of 52W Price Trend**

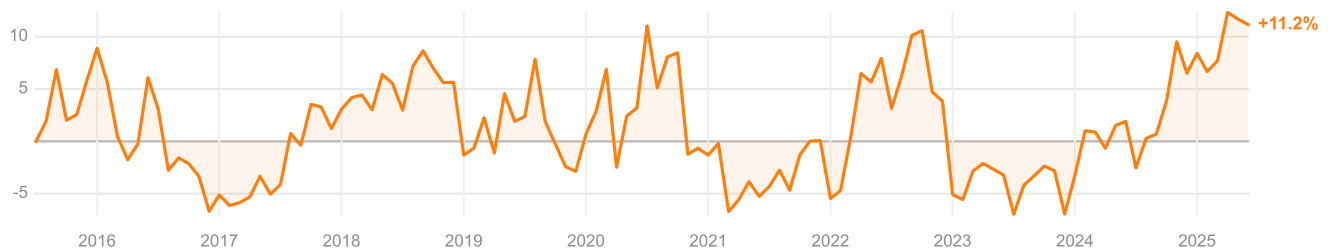
Price Momentum CAGR +11.0%

015 ● **15/36 Week Price Ratio**

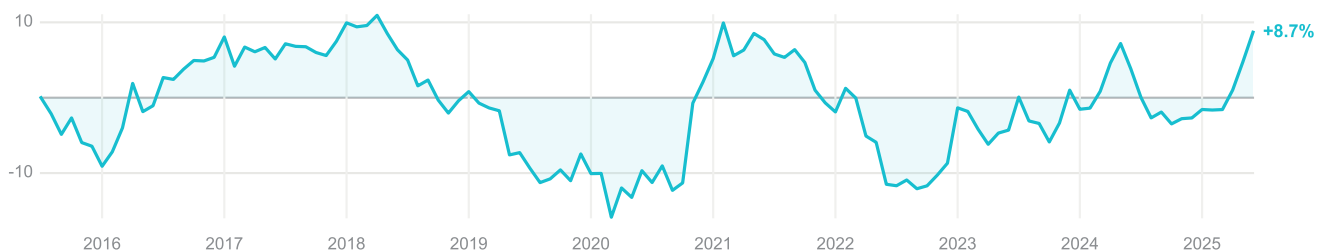
Price Momentum CAGR +10.9%

016 ● **4W to 52W Price Oscillator**

Price Momentum CAGR +10.8%

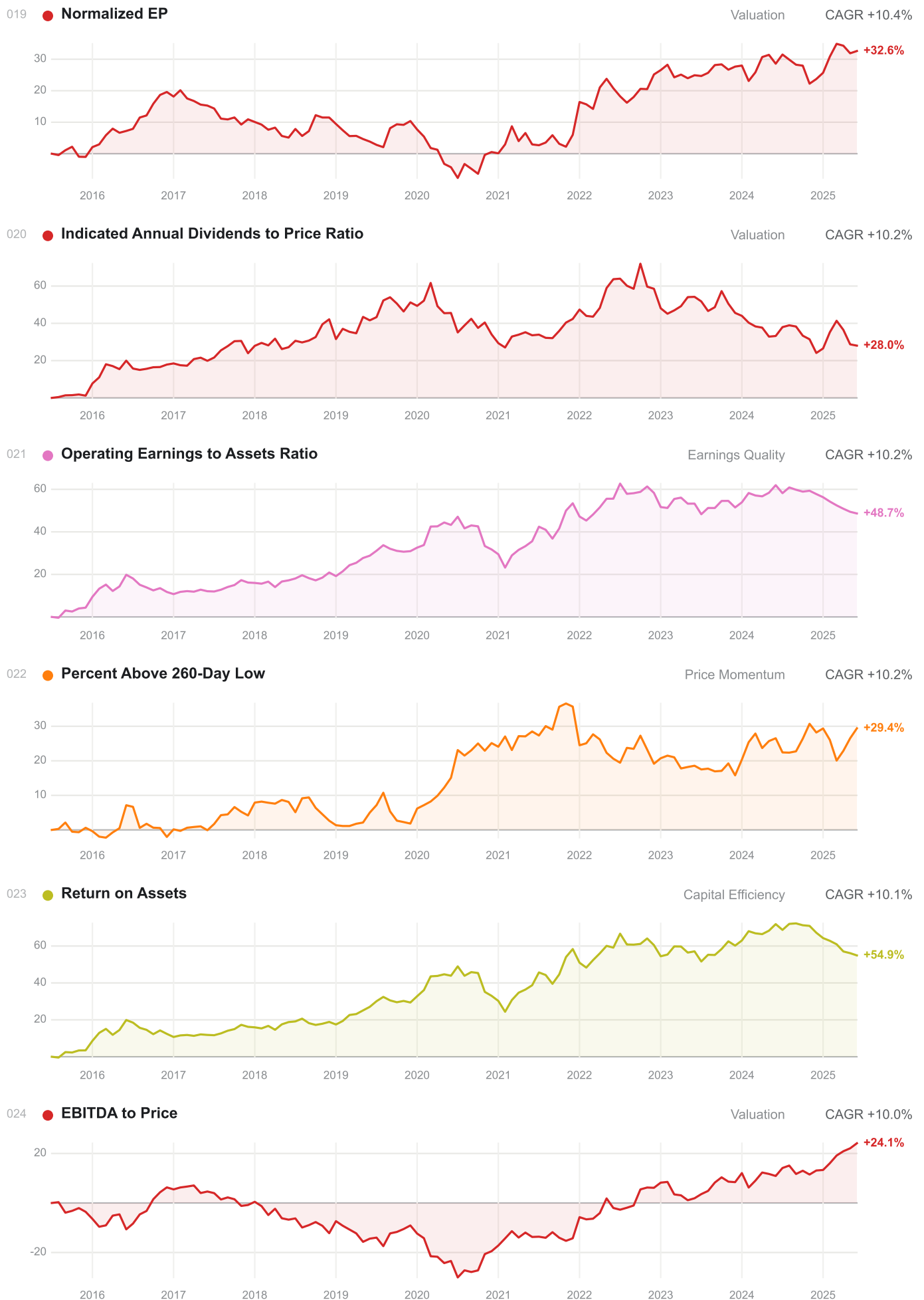
017 ● **Analyst Dispersion for FY2 EPS**

Analyst Expectations CAGR +10.7%

018 ● **Book to Price**

Valuation CAGR +10.6%

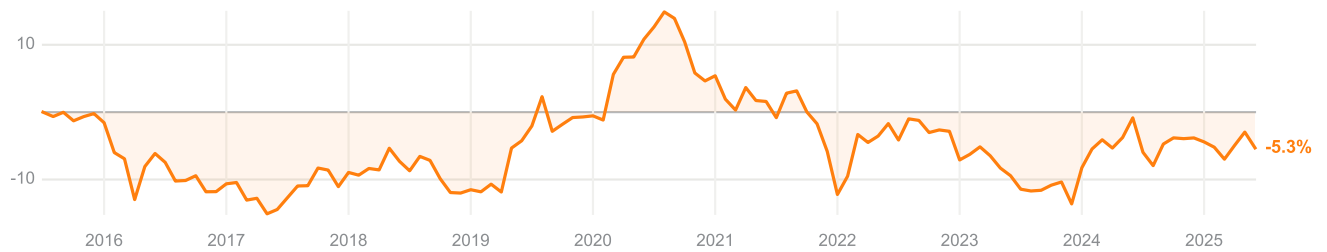




025 ● **Stability of Change in Number of Holders**

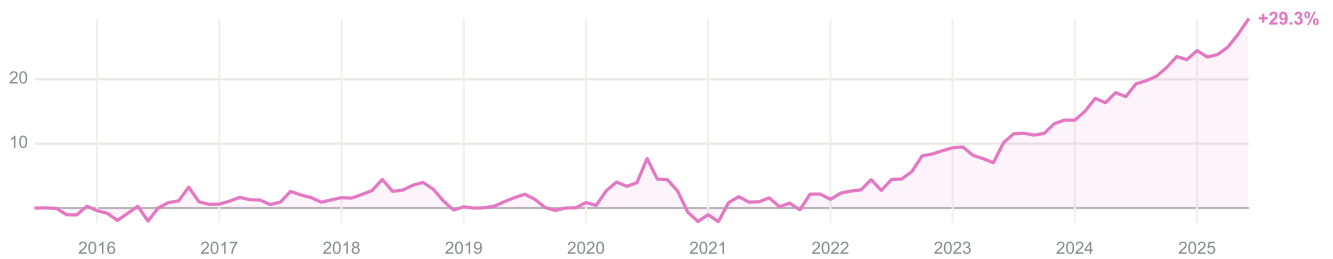
Price Momentum

CAGR +10.0%

026 ● **Unexpected Inventory Change**

Earnings Quality

CAGR +9.9%

027 ● **Log of Total Last Quarter Assets**

Size

CAGR +9.8%

028 ● **Assets to Price Ratio**

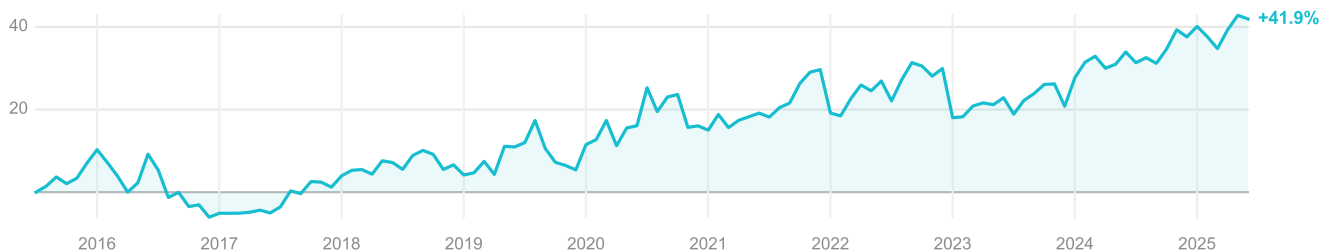
Valuation

CAGR +9.8%

029 ● **6M Chg in Target Price**

Analyst Expectations

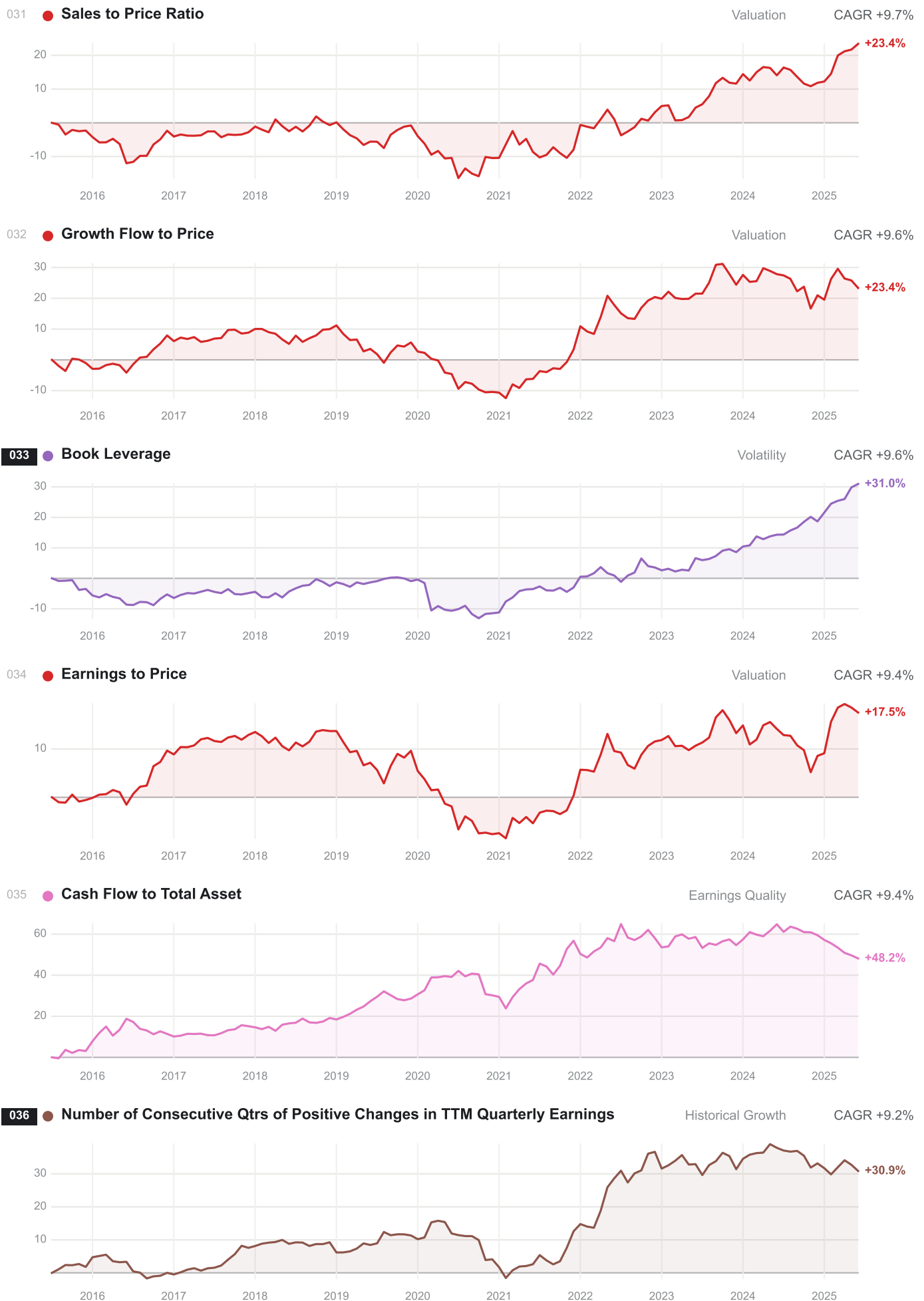
CAGR +9.7%

030 ● **Cash to Price**

Valuation

CAGR +9.7%

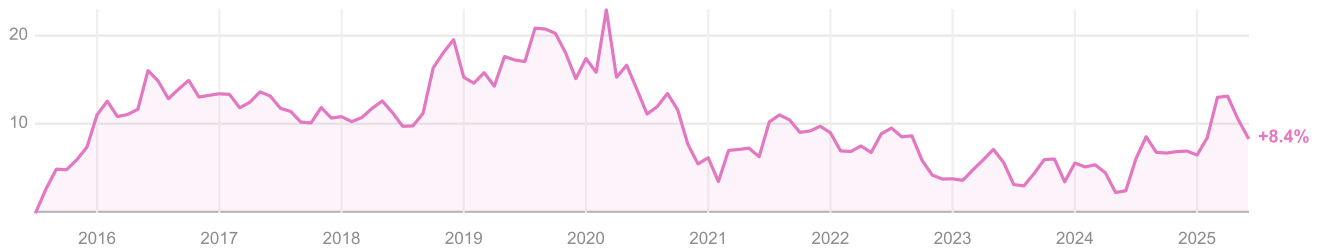




037 ● ROA 20 Qtr Standard Deviation

Earnings Quality

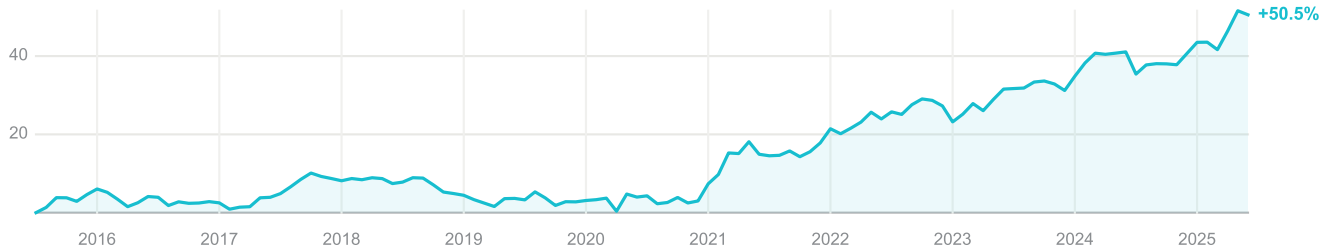
CAGR +9.2%



038 ● Adjusted 3M Revision in FY1 EPS Estimate

Analyst Expectations

CAGR +9.2%



039 ● Inverse P/E Ratio Adj for Growth and Yield

Valuation

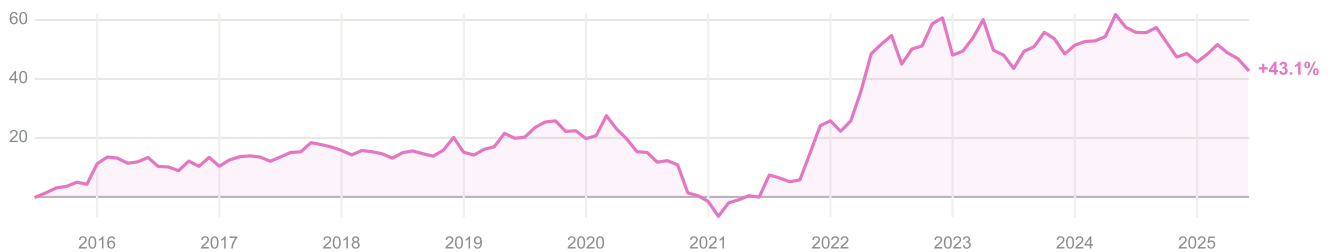
CAGR +9.1%



040 ● Trailing 12-Month Earnings/Trailing 12-Month Sales

Earnings Quality

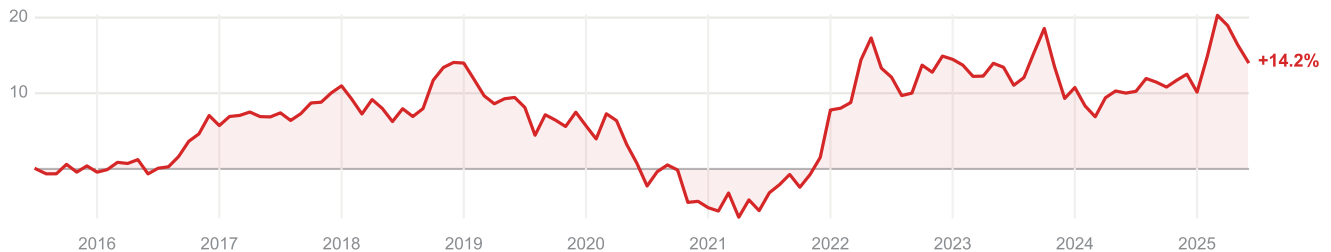
CAGR +9.1%



041 ● Net Cash Flow to Enterprise Value

Valuation

CAGR +9.1%

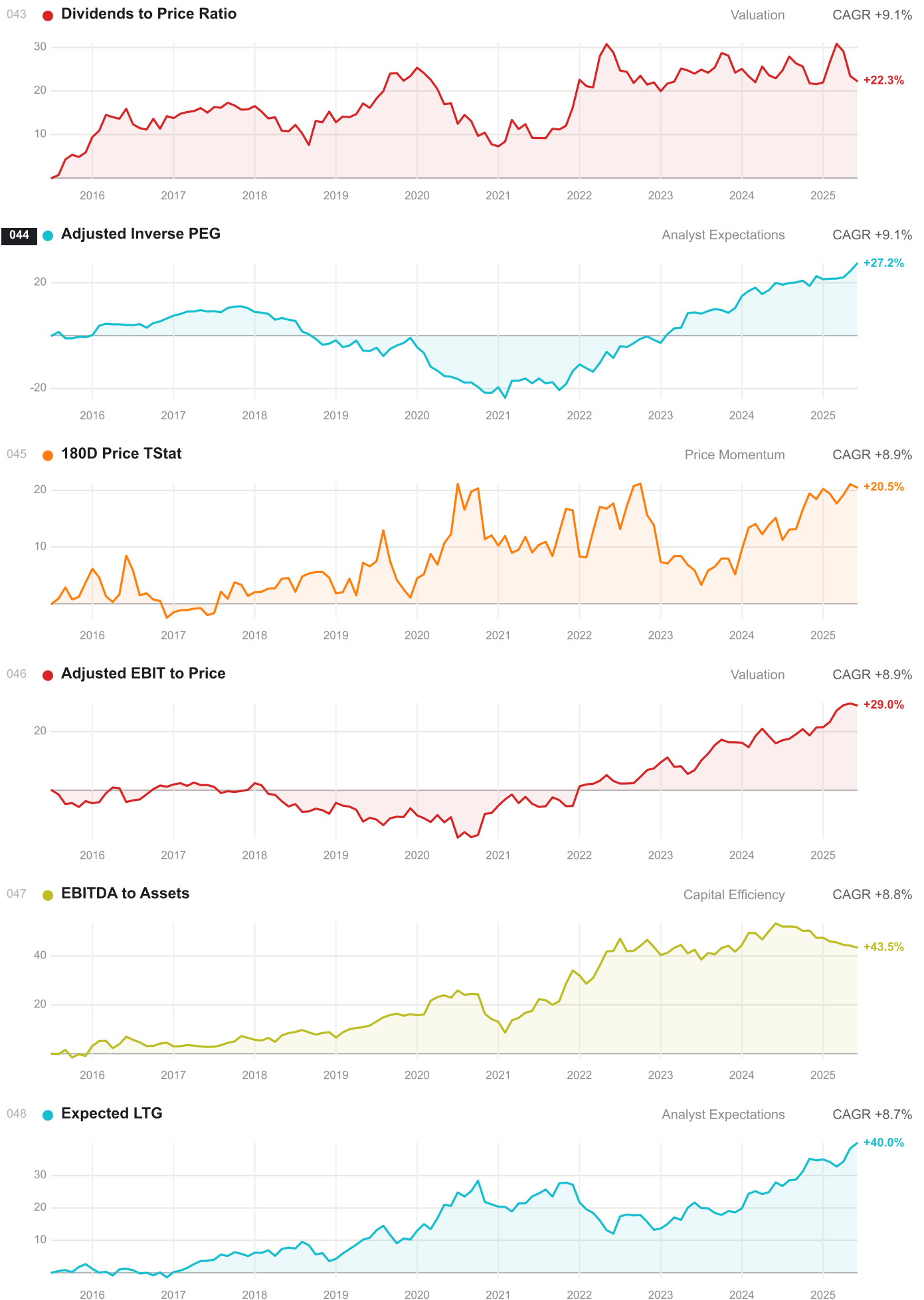


042 ● Solvency Ratio

Valuation

CAGR +9.1%

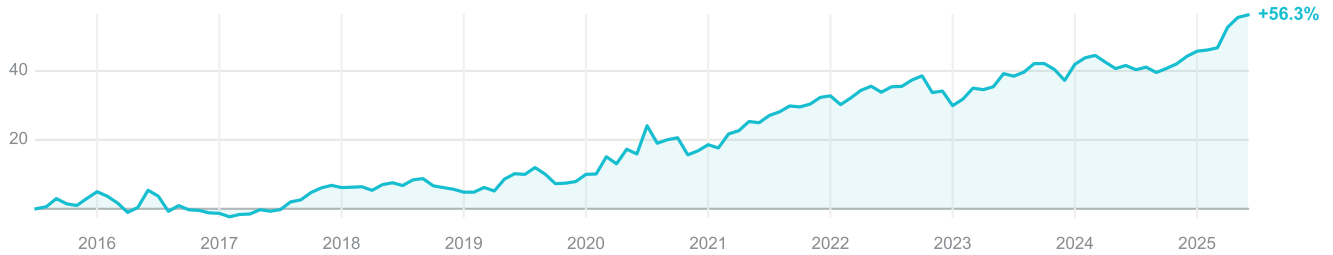




049 3M Revision in FY1 EPS Estimate

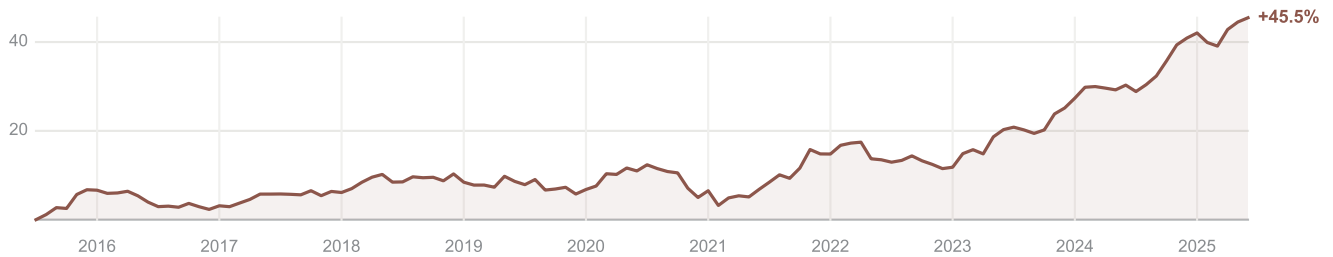
Analyst Expectations

CAGR +8.7%

**050 Adj 1Y Chg in Asset Adjusted FCF**

Historical Growth

CAGR +8.5%

**051 Sales to EV Ratio**

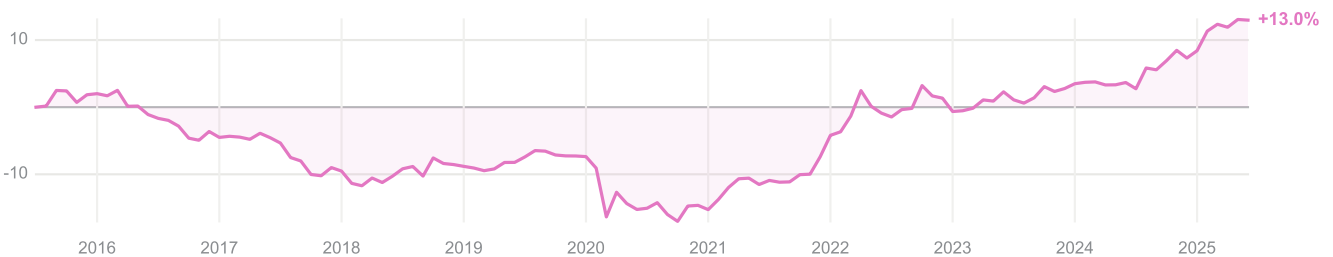
Valuation

CAGR +8.5%

**052 Working Capital to Sales**

Earnings Quality

CAGR +8.5%

**053 Adjusted Book to Enterprise Value**

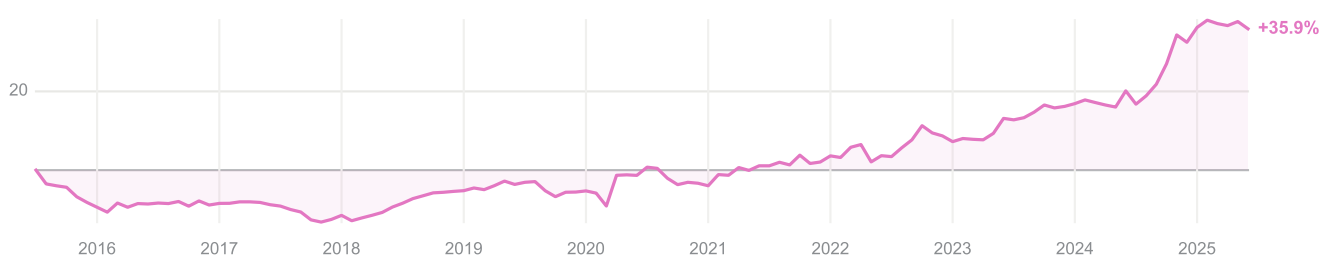
Valuation

CAGR +8.5%

**054 Balance Sheet Accruals**

Earnings Quality

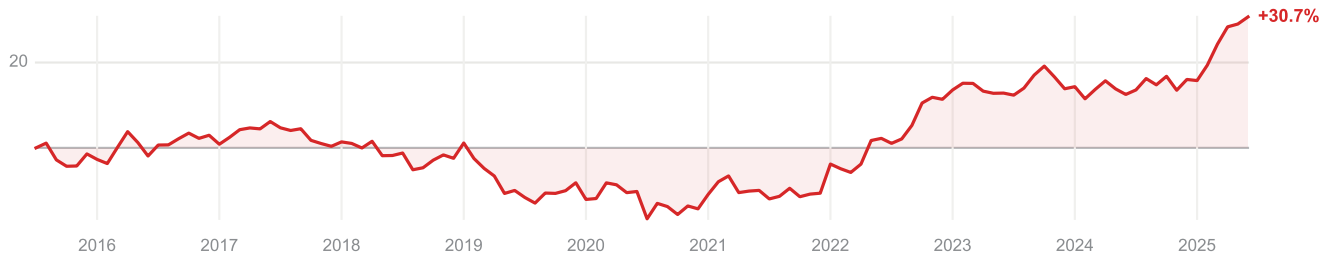
CAGR +8.4%



055 ● EBITDA to Enterprise Value

Valuation

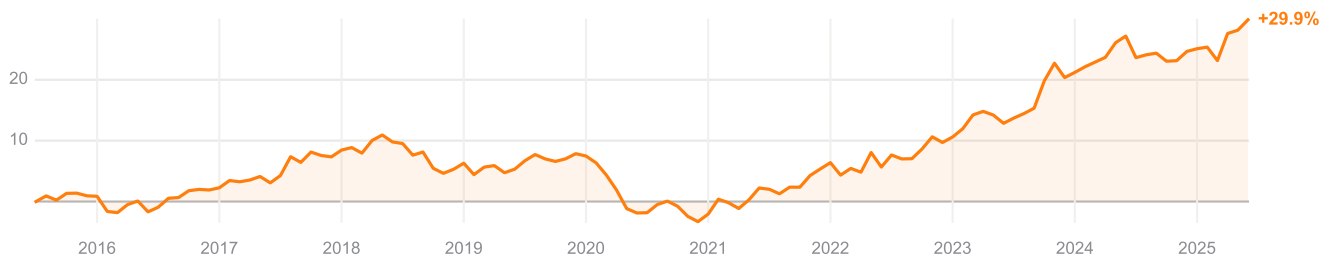
CAGR +8.4%



056 ● Short Interest Trading Volume

Price Momentum

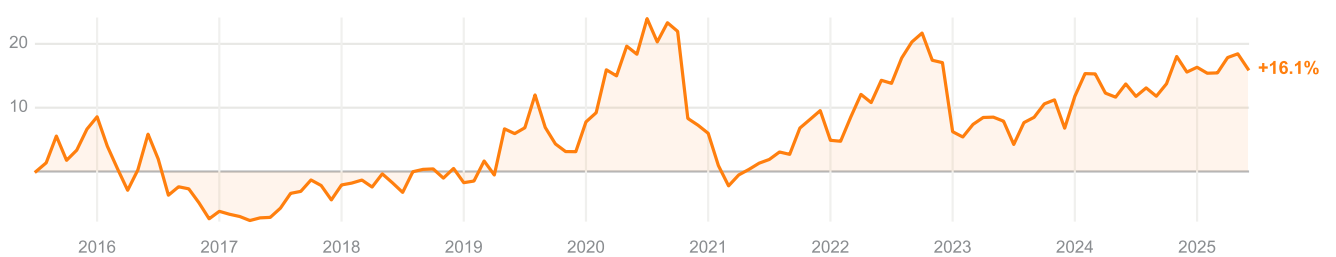
CAGR +8.4%



057 ● 12M - 1M Price Momentum

Price Momentum

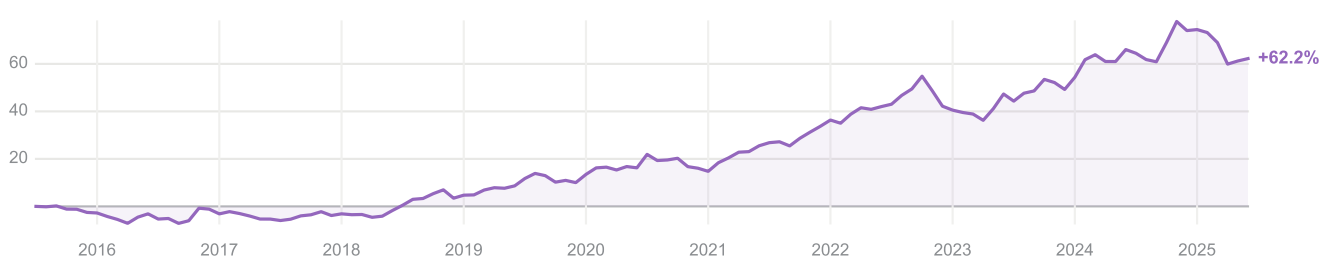
CAGR +8.3%



058 ● Amihud Illiquidity Measure

Volatility

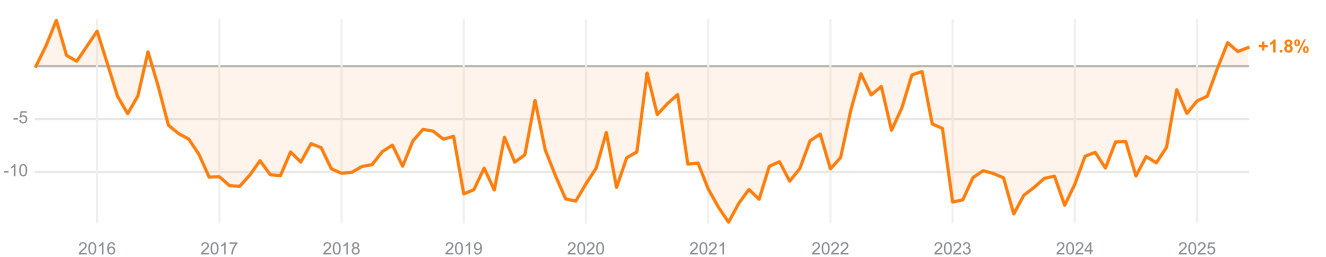
CAGR +8.3%



059 ● 52W High Low

Price Momentum

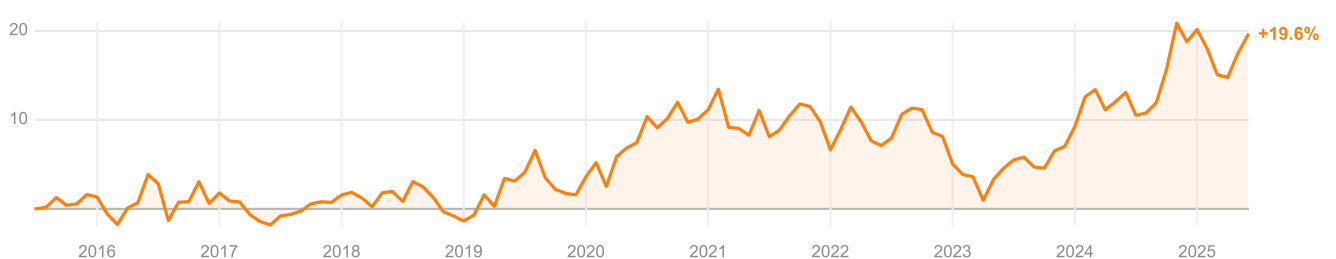
CAGR +8.3%



060 ● 52-Week Volume Price Trend with 20-Day Lag

Price Momentum

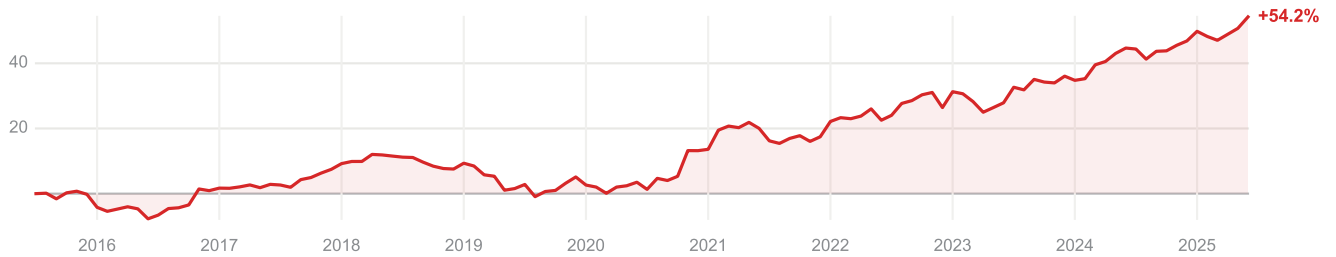
CAGR +8.2%



061 ● Cash to Adjusted Enterprise Value

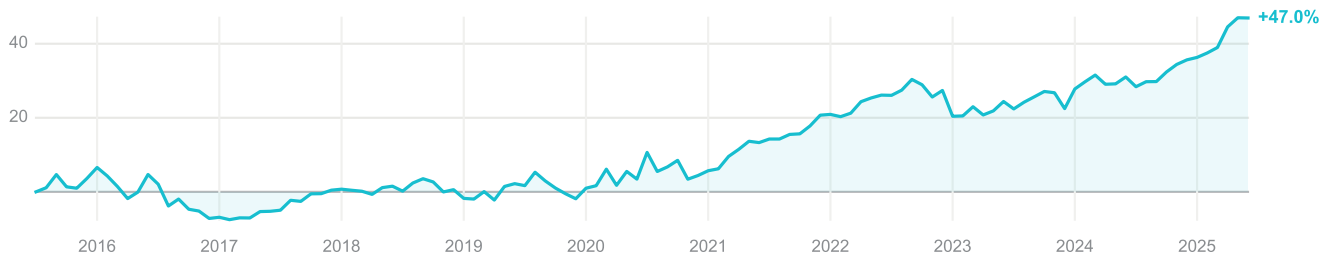
Valuation

CAGR +8.2%

**062 ● 3M Revision in FY2 EPS Estimate**

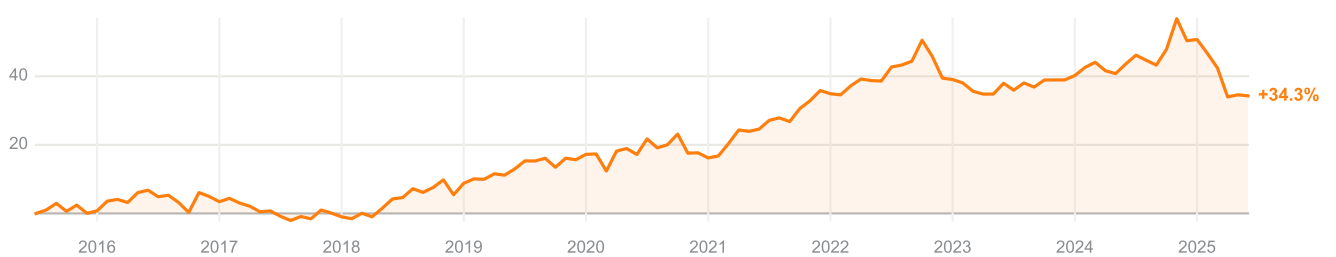
Analyst Expectations

CAGR +8.1%

**063 ● Institution Ownership Level - Active Manager**

Price Momentum

CAGR +8.1%

**064 ● Operating Cash Flow to Price**

Valuation

CAGR +8.1%

**065 ● Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1**

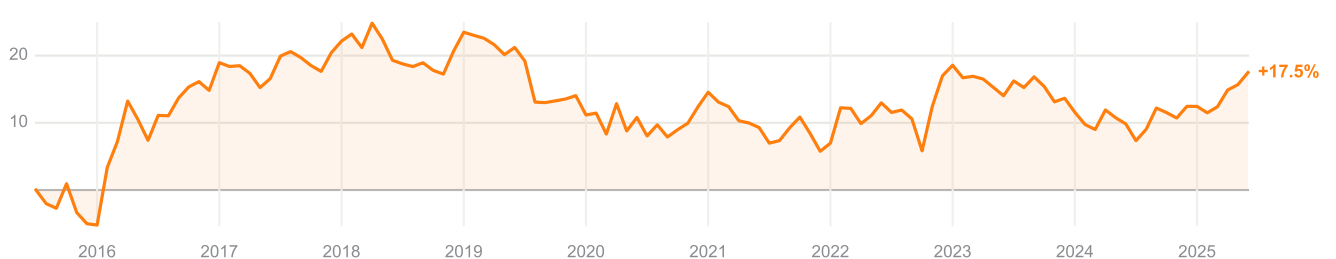
Historical Growth

CAGR +8.0%

**066 ● Log of Unadjusted Stock Price**

Price Momentum

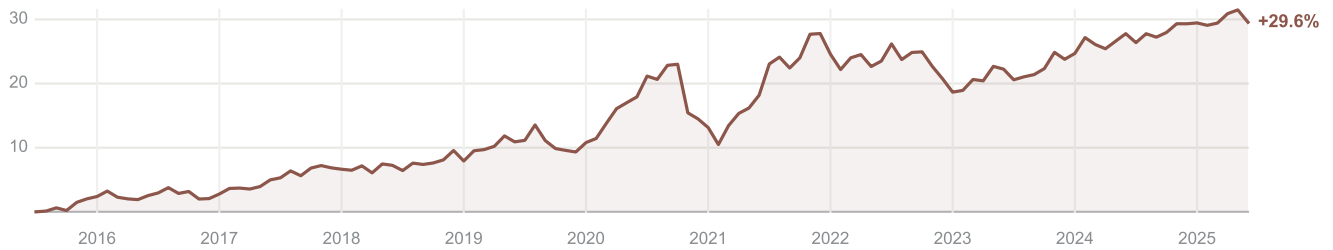
CAGR +8.0%



067 3Y Chg in Operating Cash Flow to Price

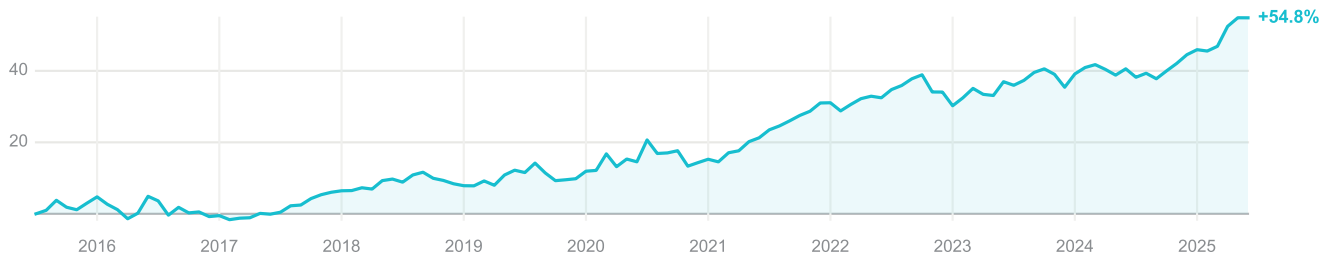
Historical Growth

CAGR +8.0%

**068 Street Revision Magnitude**

Analyst Expectations

CAGR +7.9%

**069 Cash Flow to Equity**

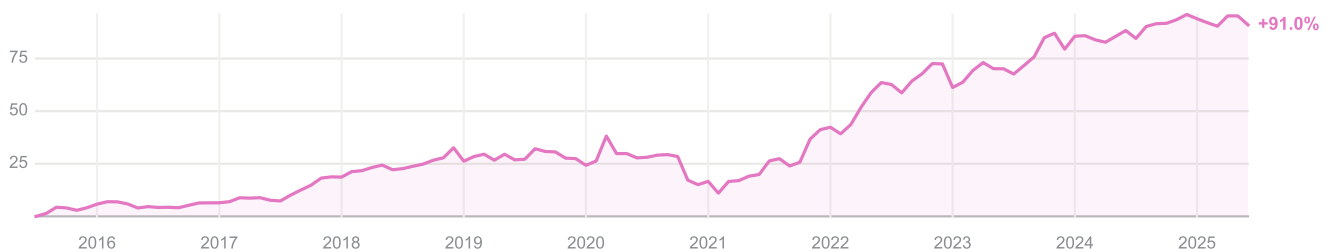
Valuation

CAGR +7.9%

**070 Operating Cash Flow to Invested Capital**

Earnings Quality

CAGR +7.9%

**071 Forward Earnings to Price**

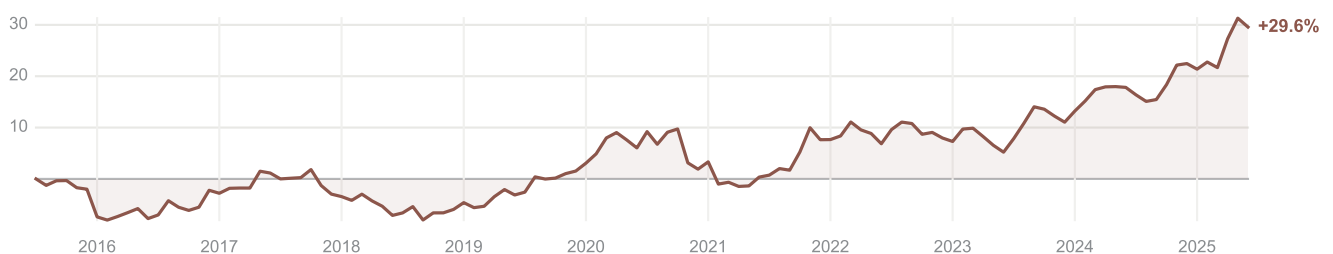
Valuation

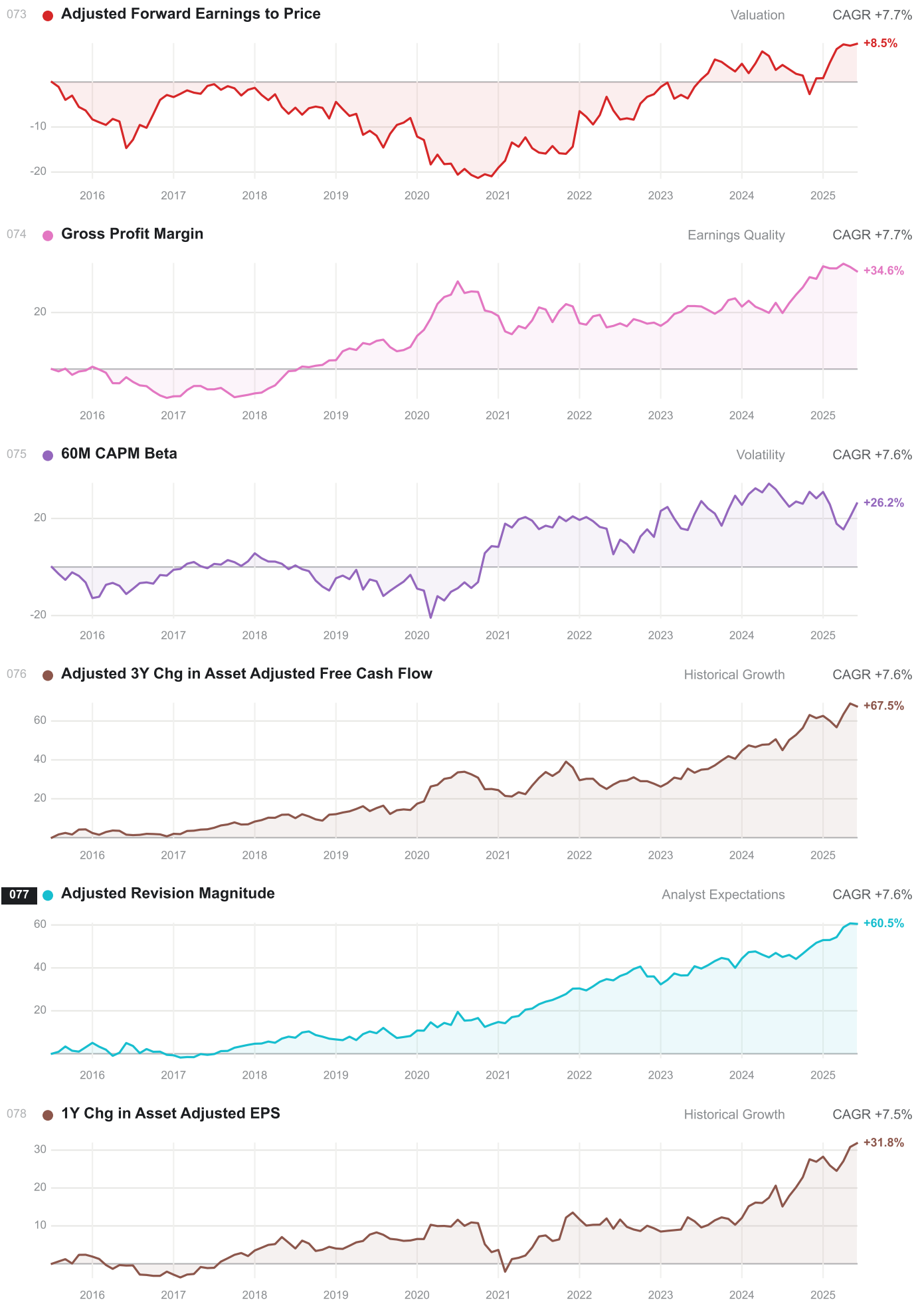
CAGR +7.9%

**072 12-Quarter Trendline in Trailing 12-Month Earnings**

Historical Growth

CAGR +7.7%





079 ● **Net Cash Flow to Price**

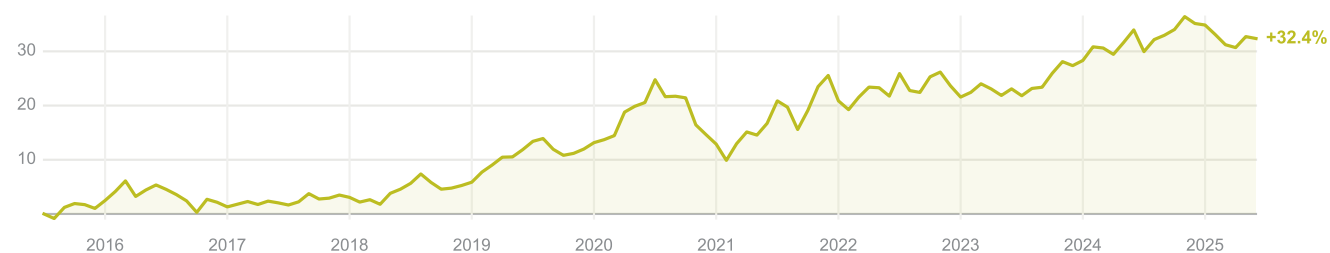
Valuation CAGR +7.5%

080 ● **Sharpe Ratio**

Price Momentum CAGR +7.3%

081 ● **Return on Equity**

Capital Efficiency CAGR +7.2%

082 ● **Operating Cash Flow to Enterprise Value**

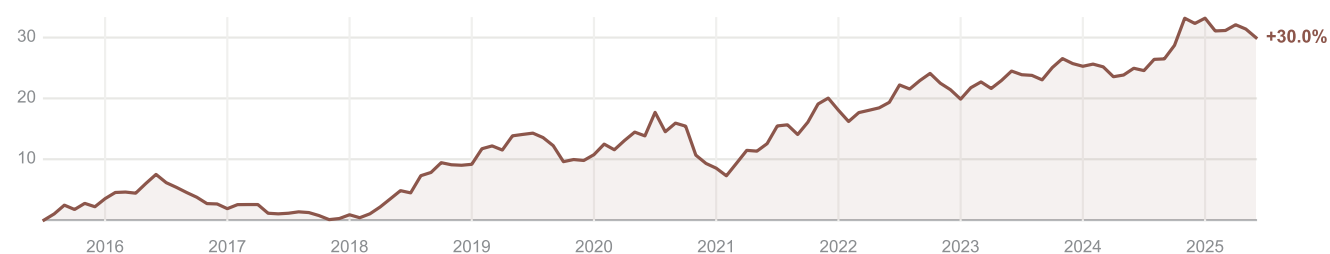
Valuation CAGR +7.2%

083 ● **1Y Chg in Operating Margin**

Historical Growth CAGR +7.2%

084 ● **1Y Chg in Free Cash Flow per Share**

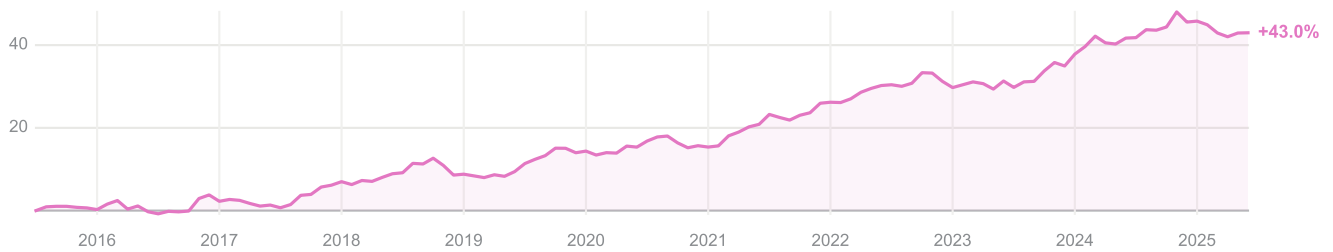
Historical Growth CAGR +7.1%



085 Equity Turnover

Earnings Quality

CAGR +7.1%

**086 1Y Chg in Asset Adjusted Cash Flow**

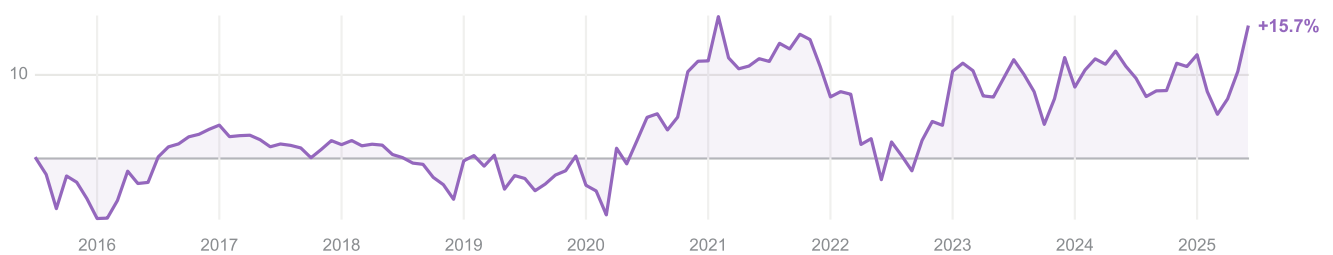
Historical Growth

CAGR +7.1%

**087 90 Day Coefficient of Variation**

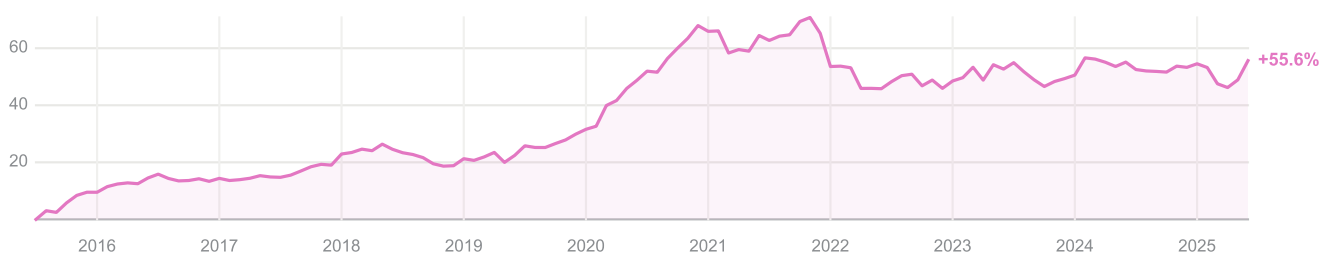
Volatility

CAGR +7.1%

**088 Working Capital to Assets Ratio**

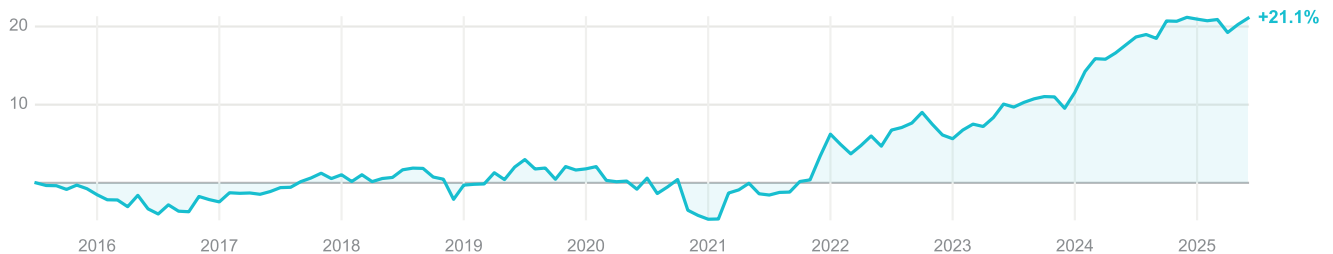
Earnings Quality

CAGR +7.0%

**089 Inverse PEG**

Analyst Expectations

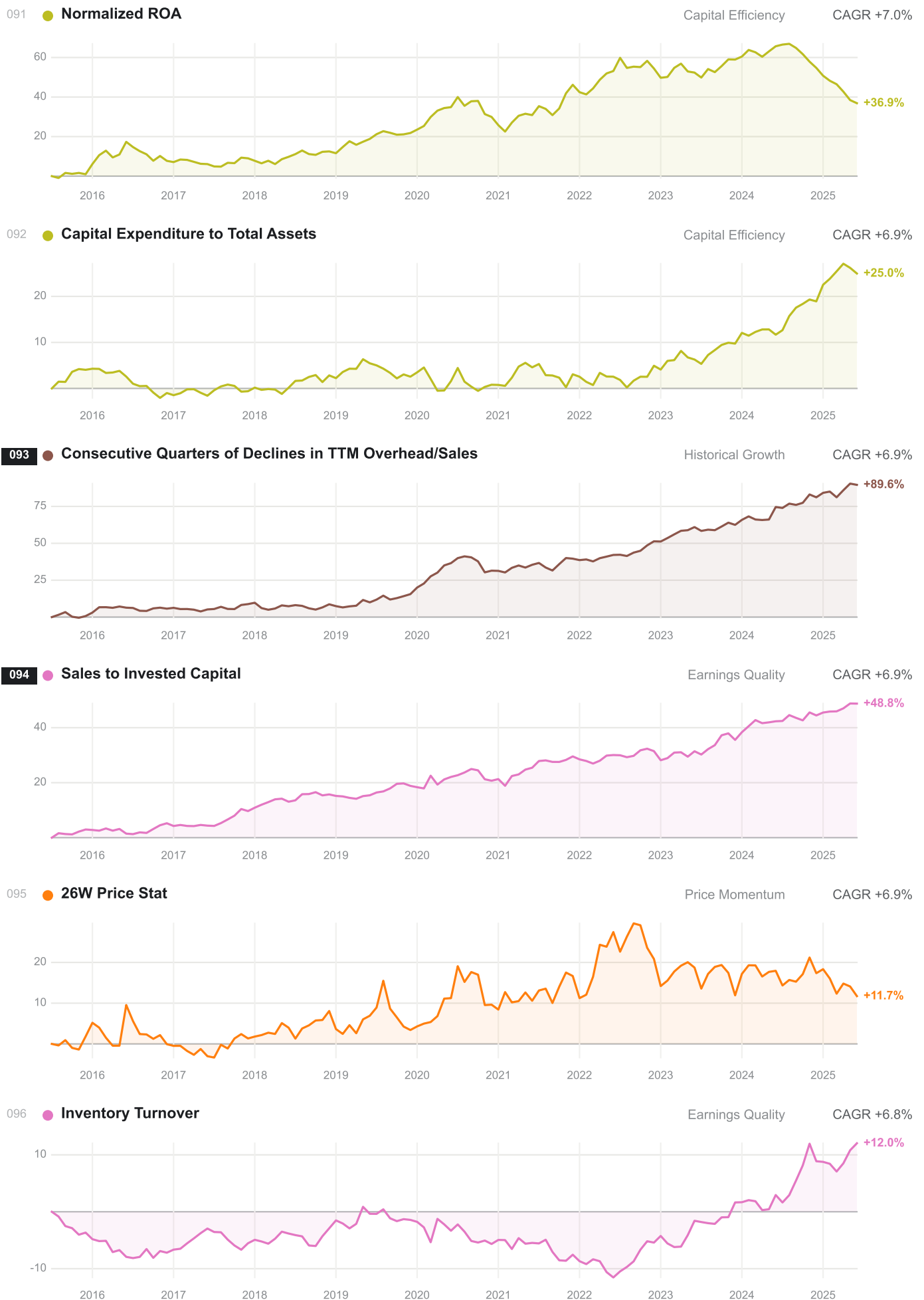
CAGR +7.0%

**090 1Y Chg in Sales to Price**

Historical Growth

CAGR +7.0%

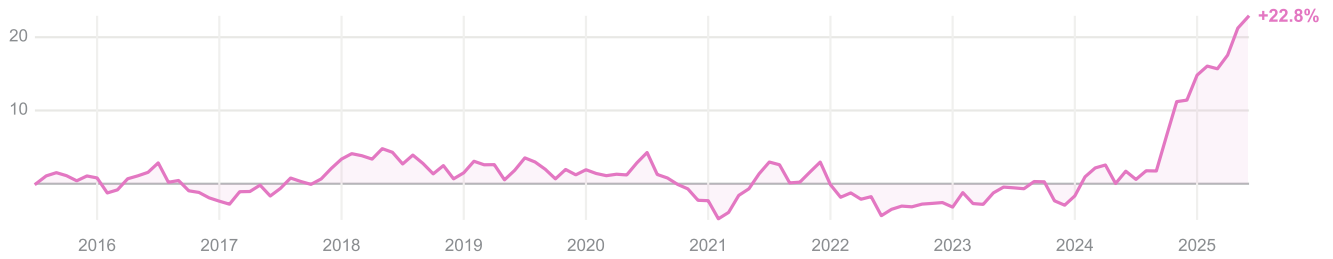




097 ● Year over Year Change in SGA to Sales

Earnings Quality

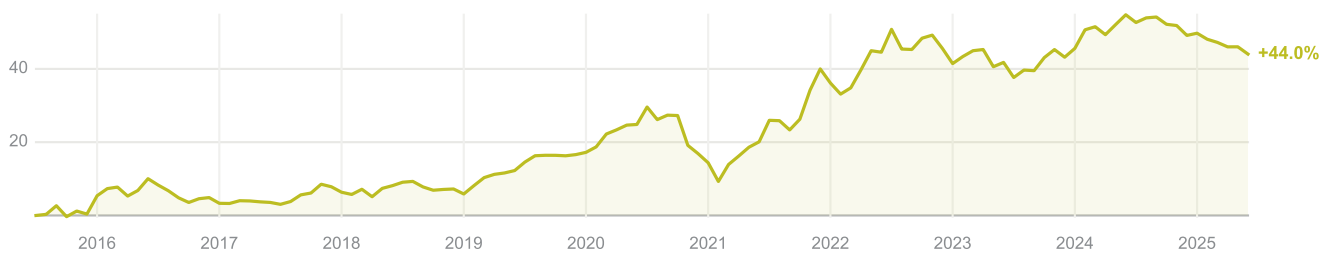
CAGR +6.7%



098 ● EBIT to Assets

Capital Efficiency

CAGR +6.7%



099 ● 1Y Chg in Earnings to Price

Historical Growth

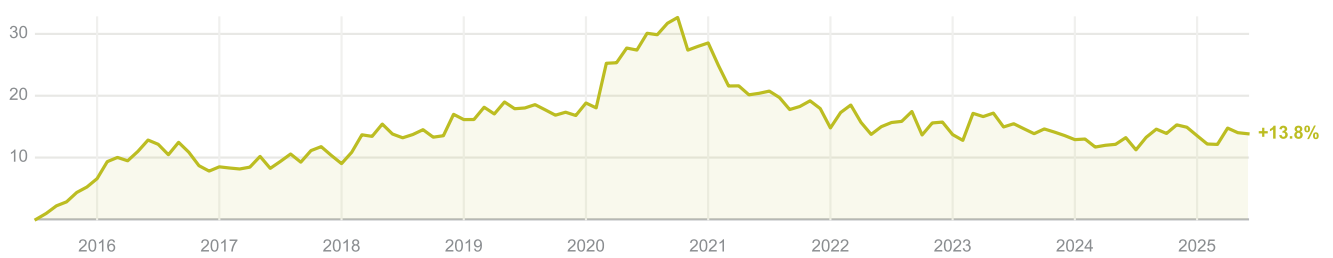
CAGR +6.7%



100 ● Long Term Debt to Equity Ratio

Capital Efficiency

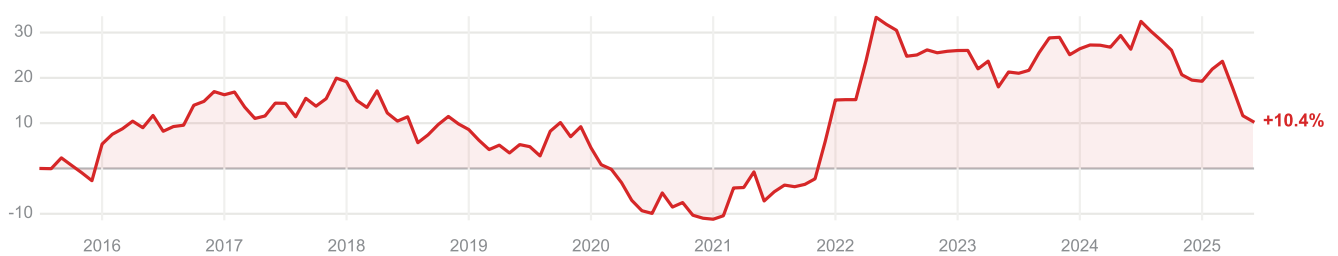
CAGR +6.7%



101 ● Adj. EBITDA to Enterprise Value

Valuation

CAGR +6.6%



102 ● Adj 1Y Chg in Earnings per Share

Historical Growth

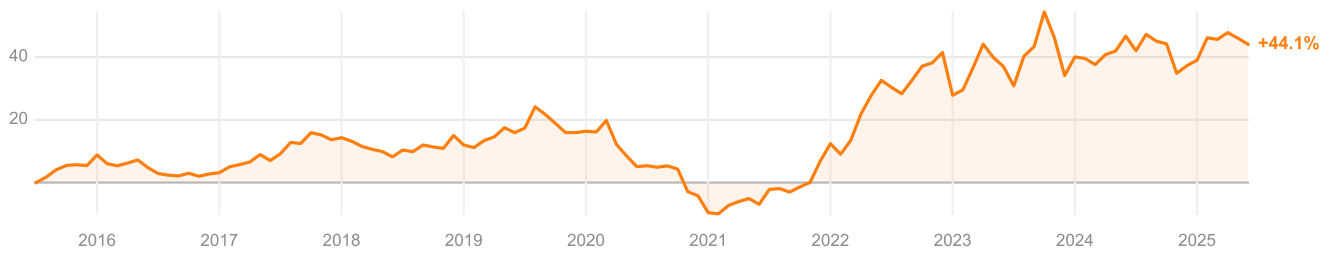
CAGR +6.6%



103 ● **Short Interest Ratio**

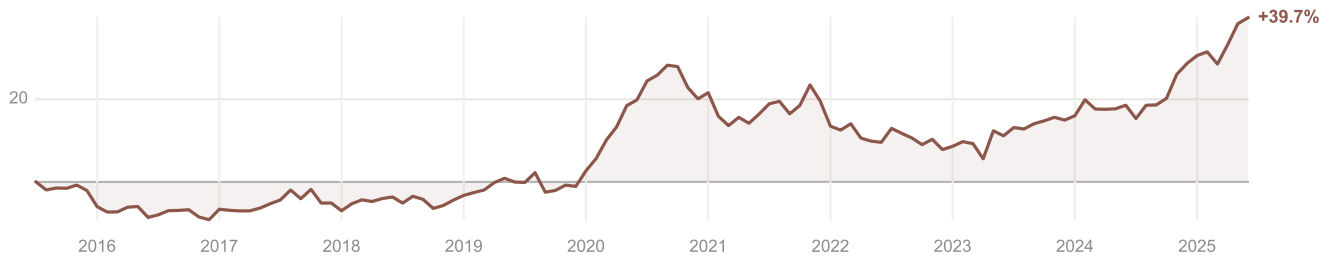
Price Momentum

CAGR +6.6%

104 ● **12-Quarter Trendline in Trailing 12-Month Cash Flows**

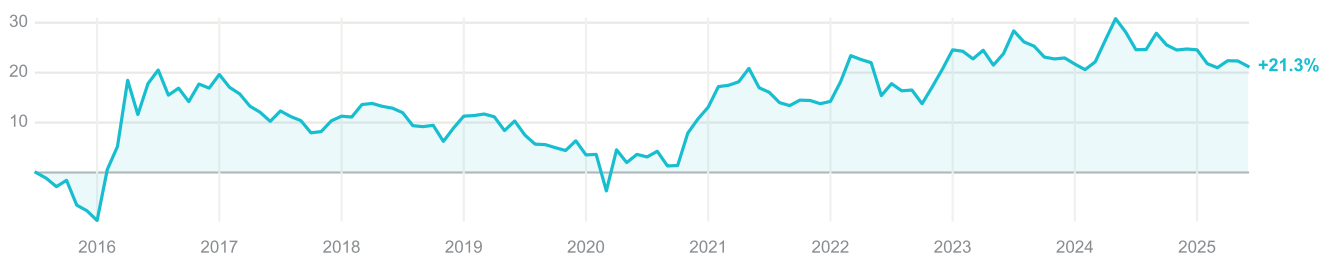
Historical Growth

CAGR +6.5%

105 ● **Analyst Dispersion for FY1 EPS**

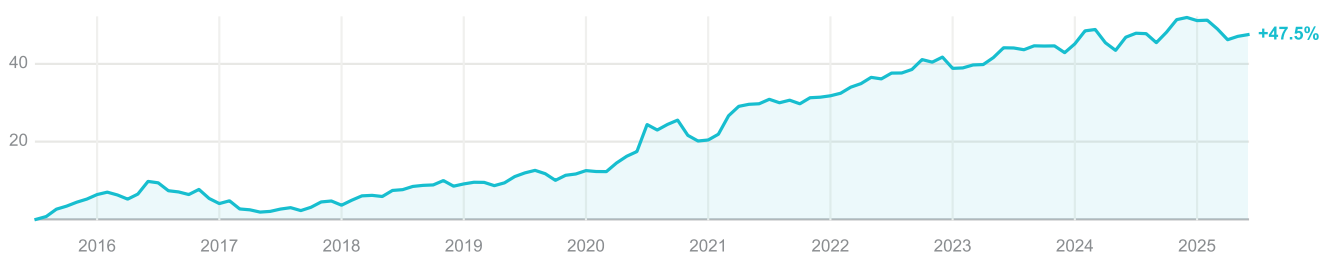
Analyst Expectations

CAGR +6.5%

106 ● **Earnings Surprise**

Analyst Expectations

CAGR +6.4%

107 ● **Operating Cash Flow Ratio**

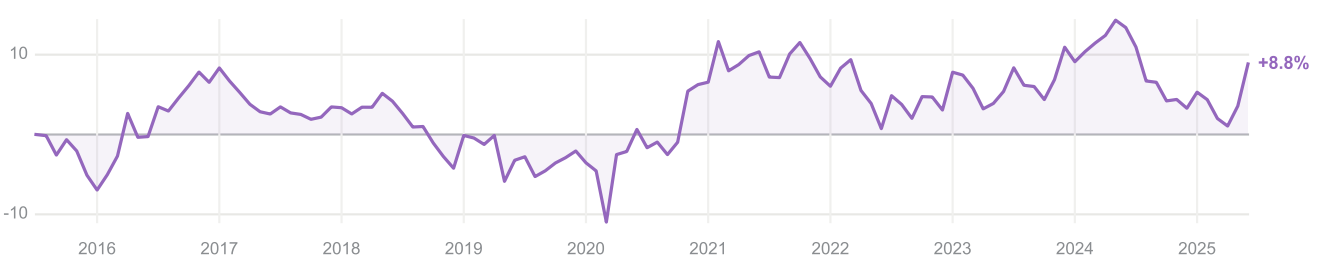
Capital Efficiency

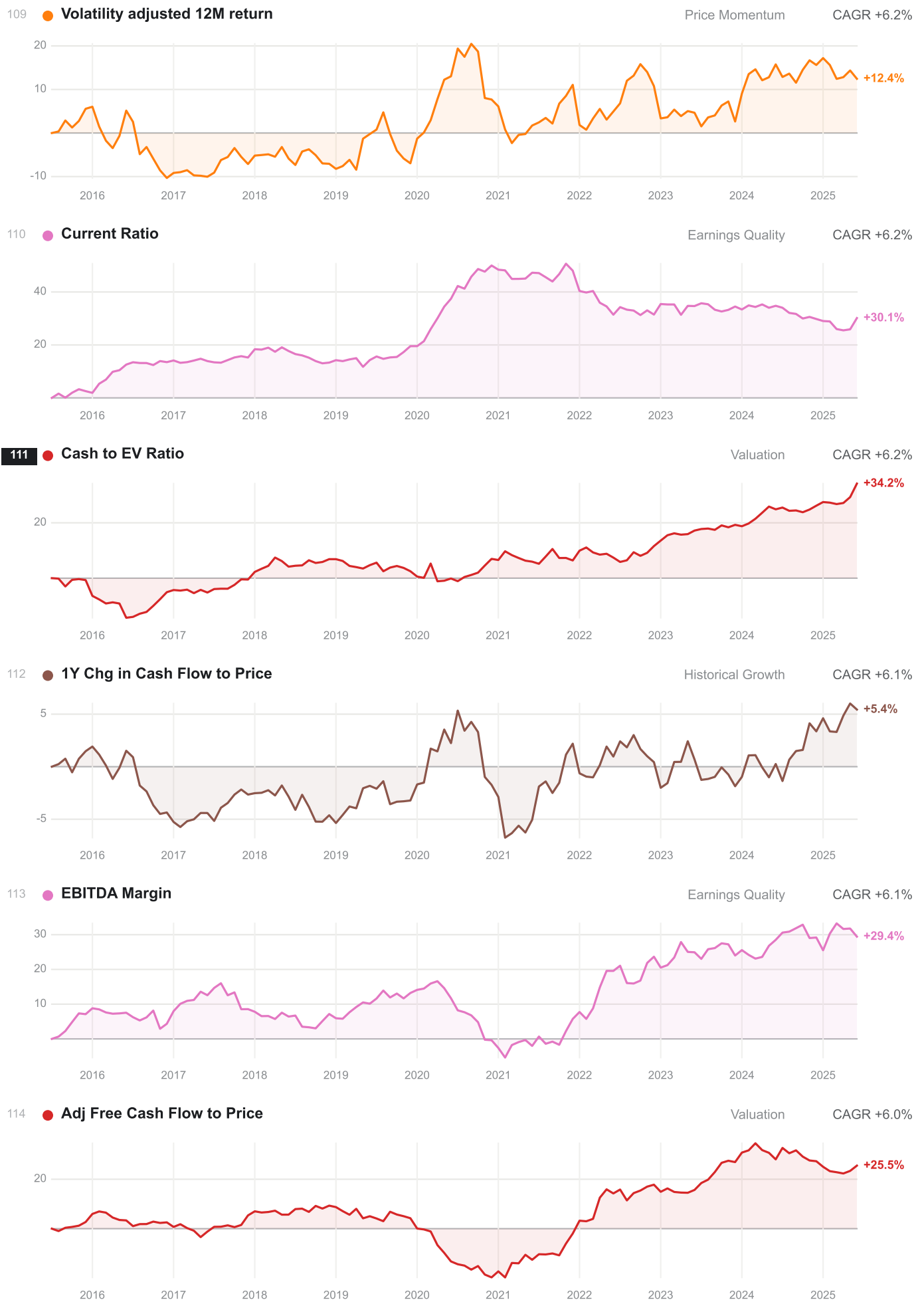
CAGR +6.3%

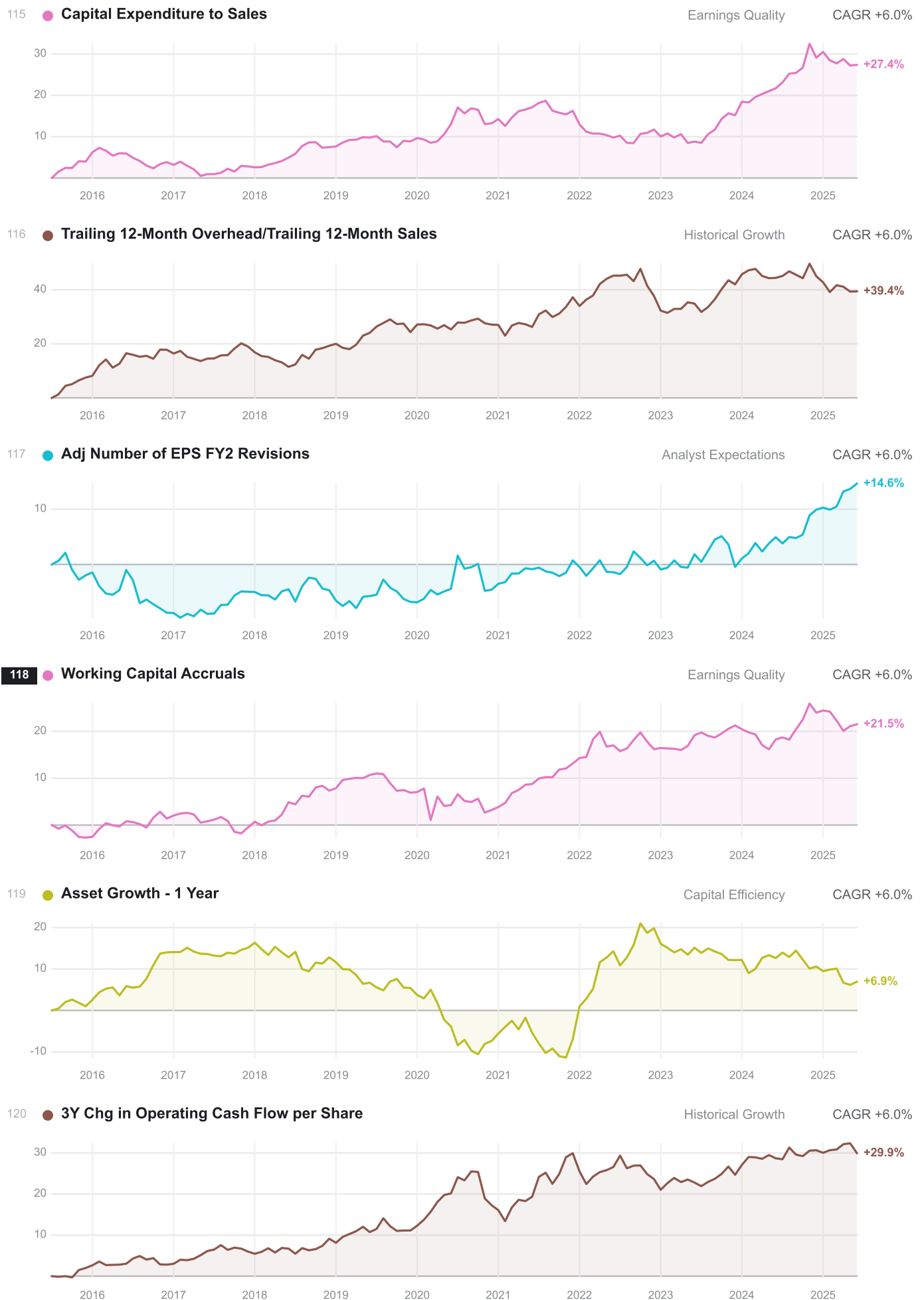
108 ● **1M Realized Price Volatility**

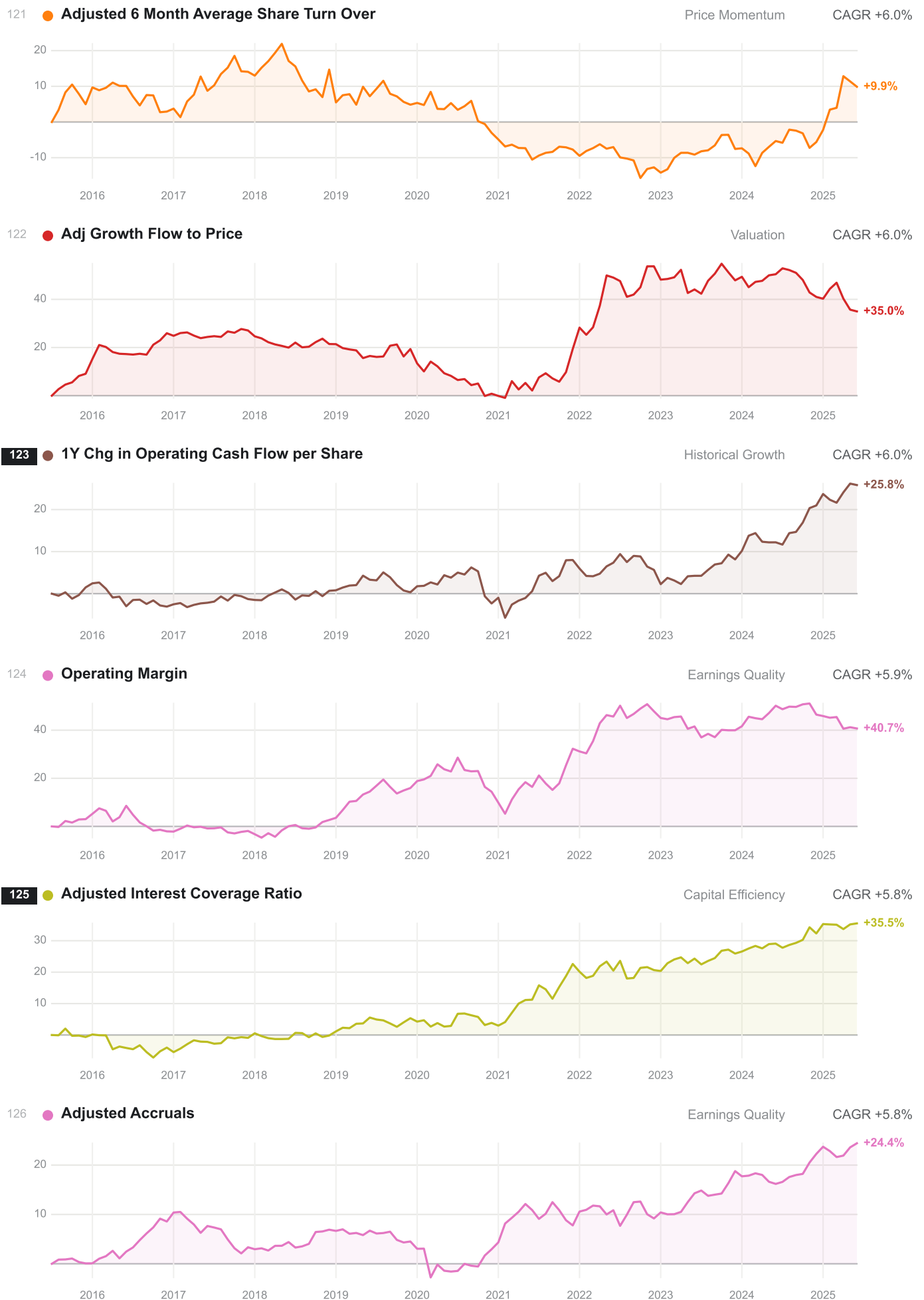
Volatility

CAGR +6.2%









127 ● 3Y Chg in Free Cash Flow to Price

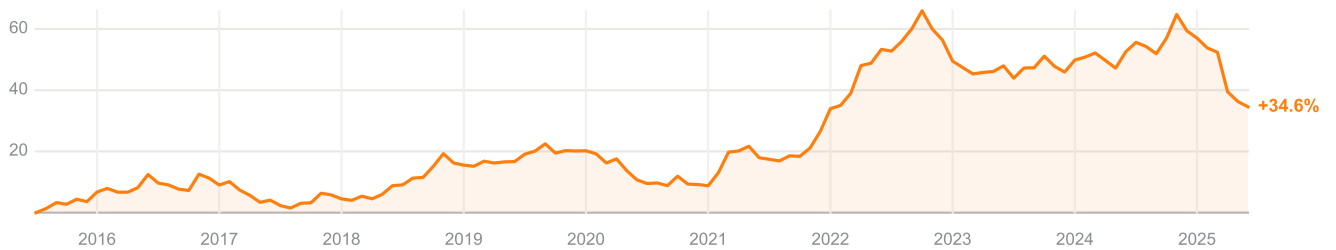
Historical Growth

CAGR +5.7%

**128 ● Ownership Breadth Stability of Institution Ownership by Buyers**

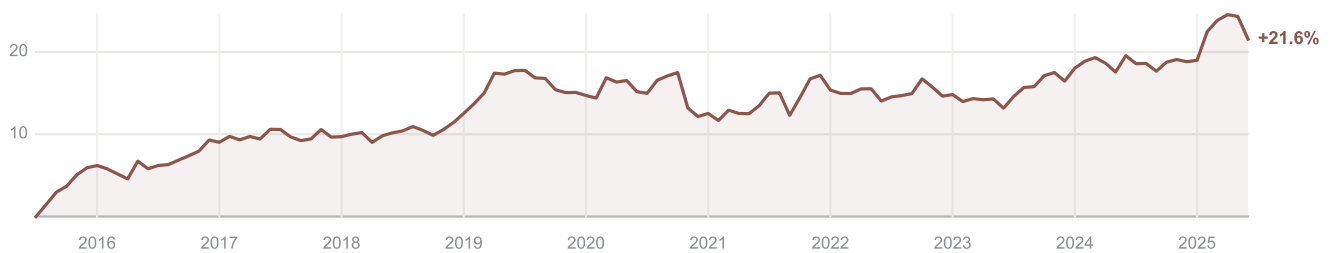
Price Momentum

CAGR +5.7%

**129 ● Dividend per Share Growth**

Historical Growth

CAGR +5.7%

**130 ● Accrual Ratio - Balance Sheet**

Earnings Quality

CAGR +5.6%

**131 ● Long Term Debt to Assets Ratio**

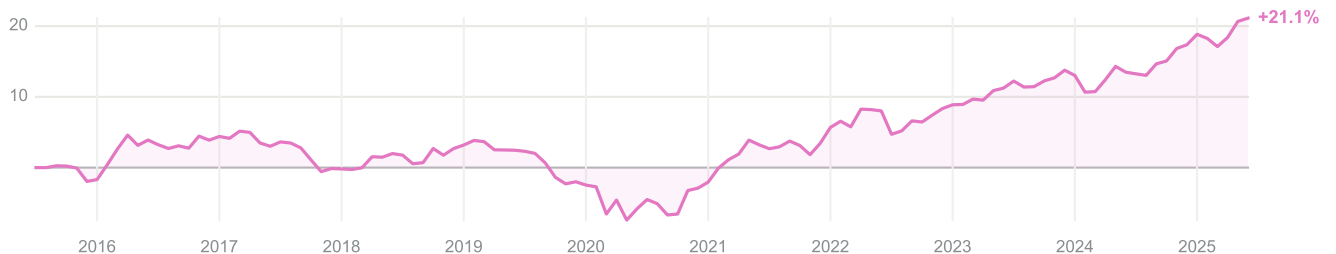
Capital Efficiency

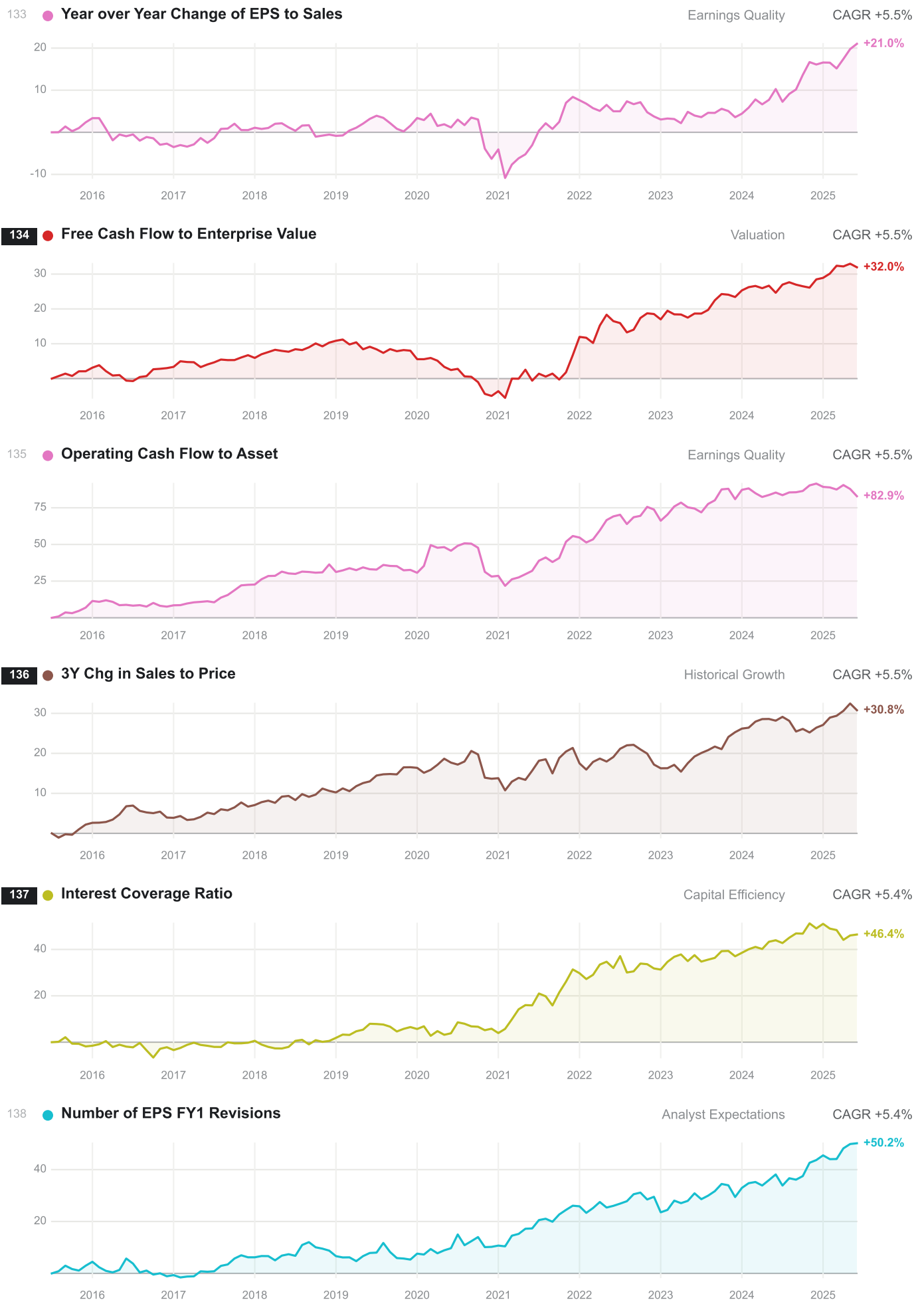
CAGR +5.6%

**132 ● Accrual Ratio - Cash Flows**

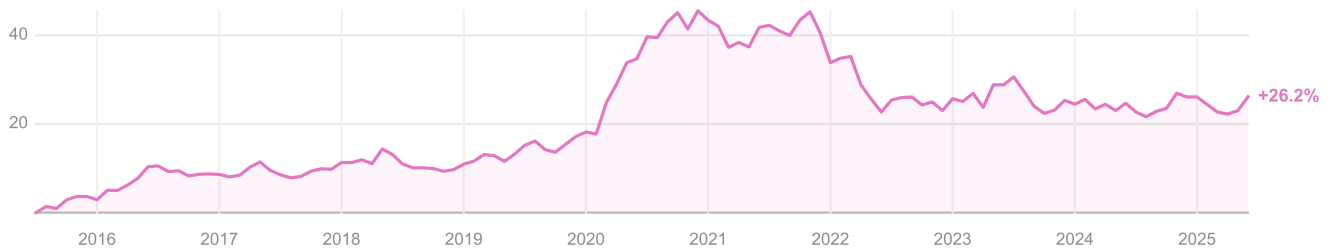
Earnings Quality

CAGR +5.6%



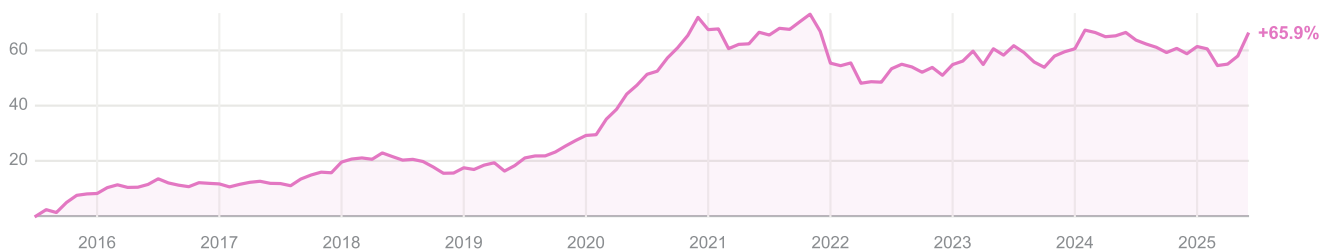


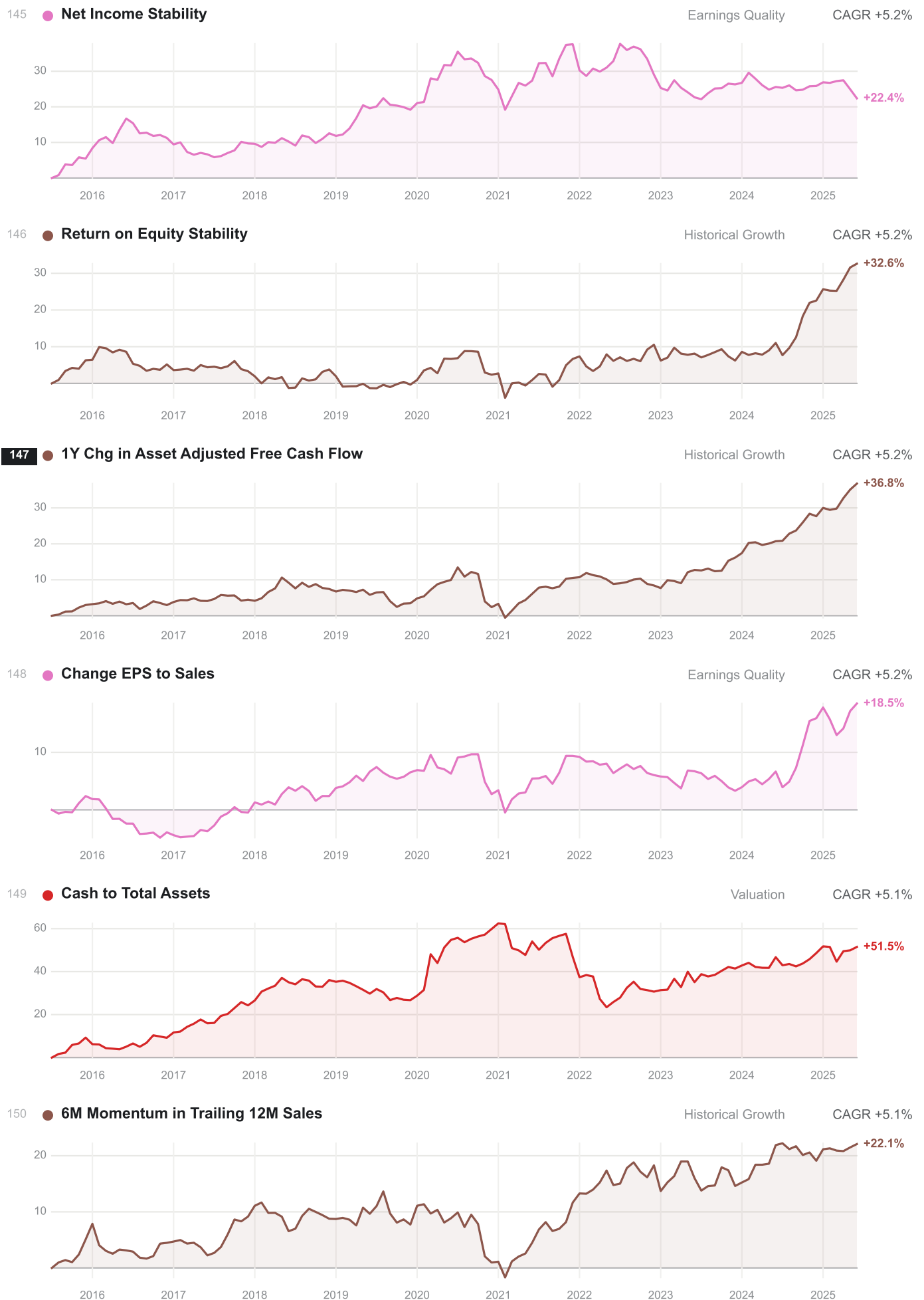
139 ● Num of Quarters of Declines in Receivables and Inventories/TTM Sales
Historical Growth CAGR +5.4%

140 ● Quick Ratio
Earnings Quality CAGR +5.3%

141 ● 1Y Chg in Asset Adjusted Operating Cash Flow
Historical Growth CAGR +5.3%

142 ● Debt to Assets Ratio
Capital Efficiency CAGR +5.3%

143 ● Forward Free Cash Flow to Price Ratio
Valuation CAGR +5.3%

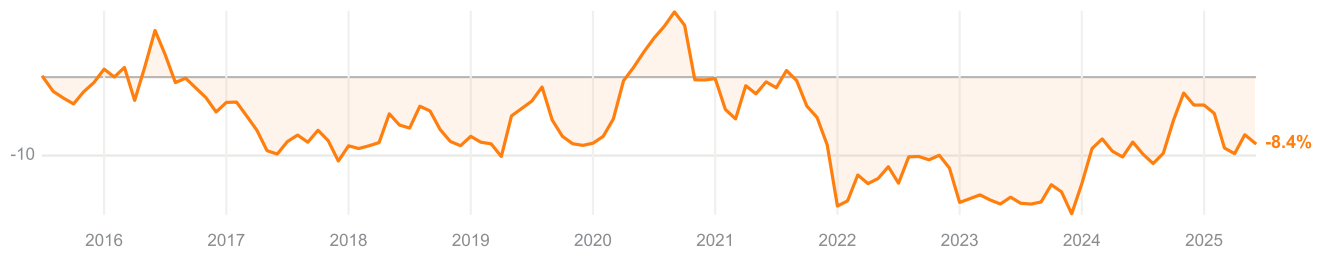
144 ● Working Capital to Total Assets
Earnings Quality CAGR +5.2%




151 ● **Change in Number of Holders**

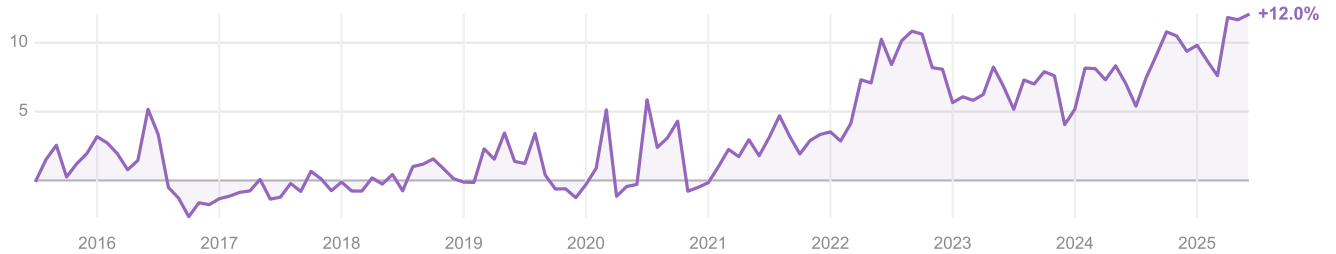
Price Momentum

CAGR +5.1%

152 ● **1Y Chg in Amihud**

Volatility

CAGR +5.0%

153 ● **24M Residual Return Variance**

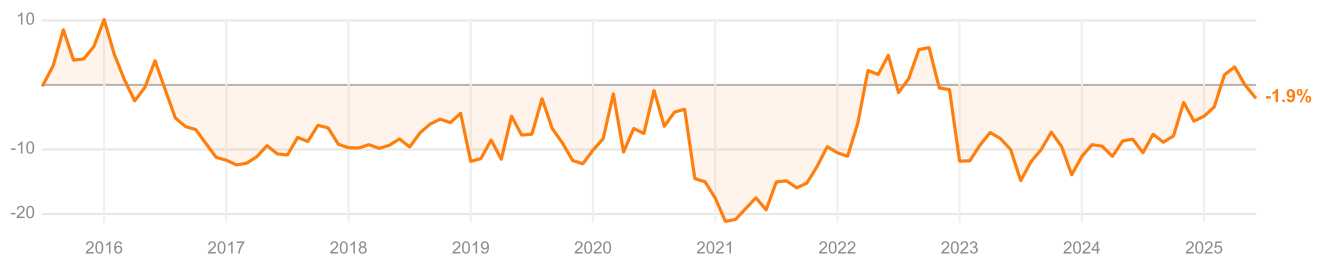
Price Momentum

CAGR +5.0%

154 ● **Closing Price to 52 Week High**

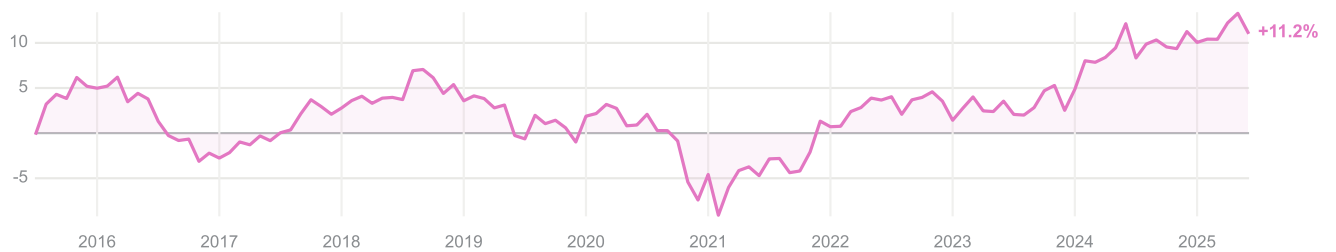
Price Momentum

CAGR +5.0%

155 ● **Free Cash Flow Stability**

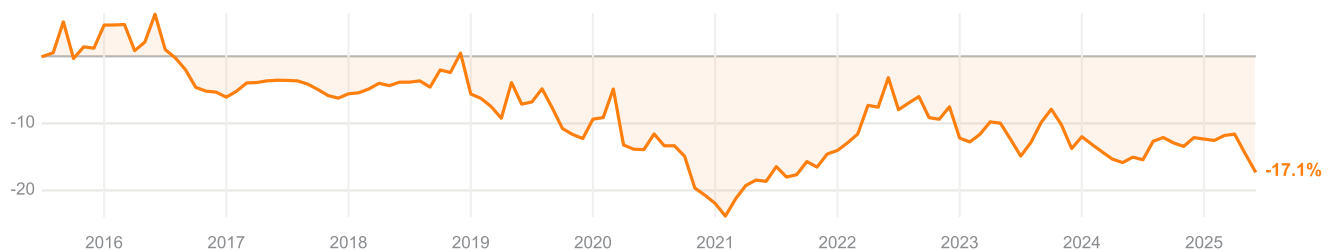
Earnings Quality

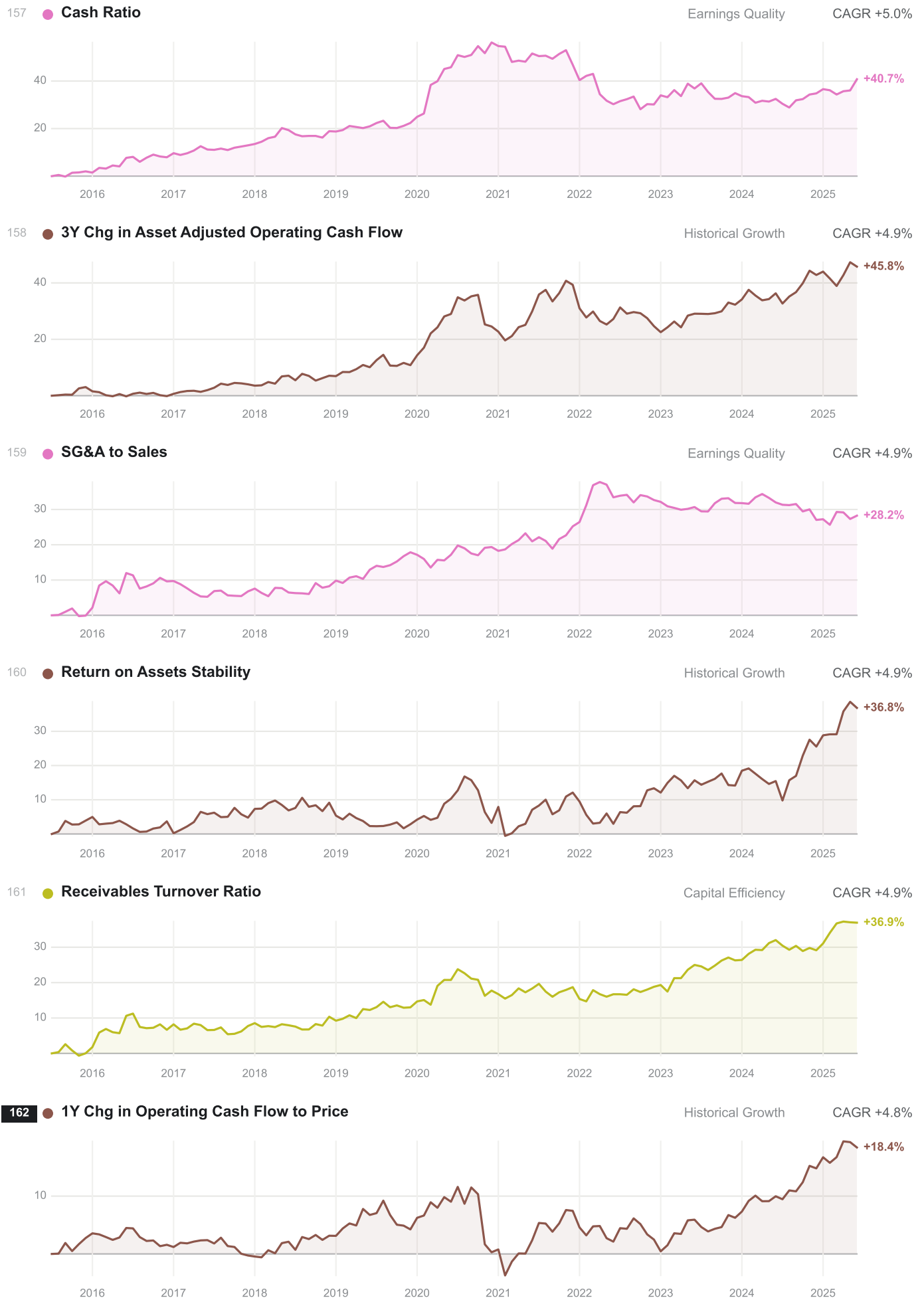
CAGR +5.0%

156 ● **30 Day Coefficient of Variation of Volume to Price**

Price Momentum

CAGR +5.0%

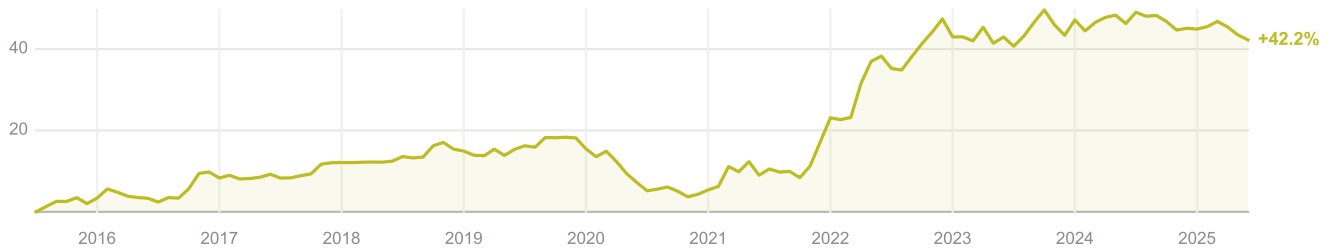




163 ● 1Y Chg in Shares Outstanding

Capital Efficiency

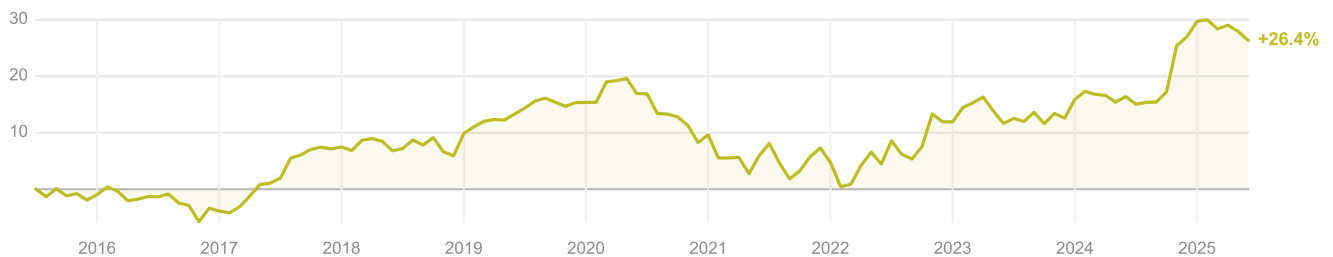
CAGR +4.8%



164 ● 1Y Chg in Stock Buybacks

Capital Efficiency

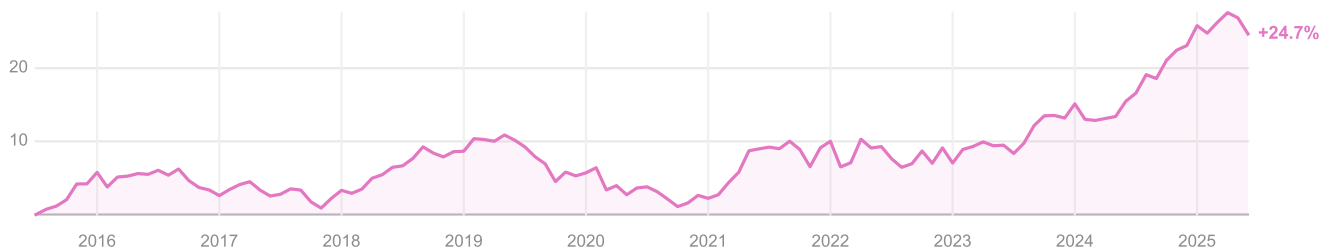
CAGR +4.8%



165 ● Depreciation to Capex Ratio

Earnings Quality

CAGR +4.8%



166 ● Free Cash Flow to Price

Valuation

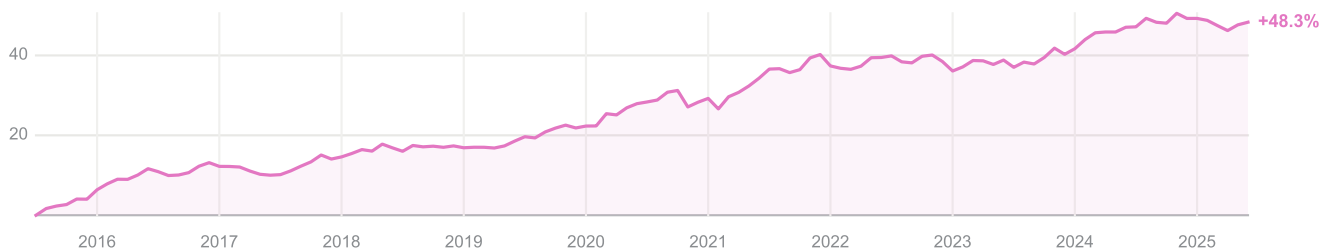
CAGR +4.7%



167 ● Asset Turnover

Earnings Quality

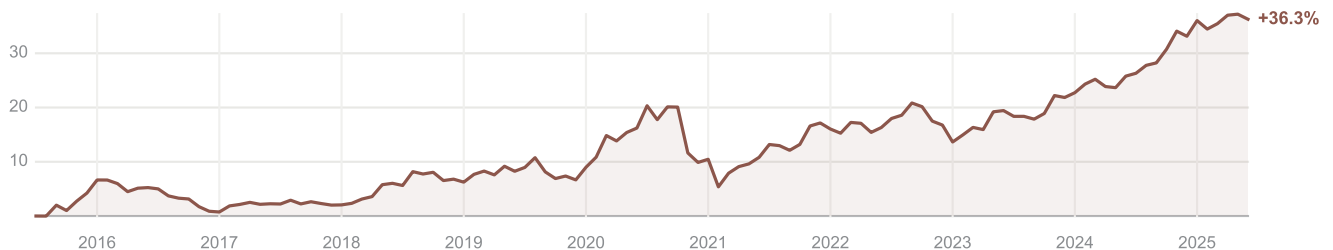
CAGR +4.7%



168 ● 1Y Chg in Free Cash Flow to Price

Historical Growth

CAGR +4.6%



169 ● **Adj Forward Free Cash Flow to Price Ratio**

Valuation

CAGR +4.6%

170 ● **EPS Stability**

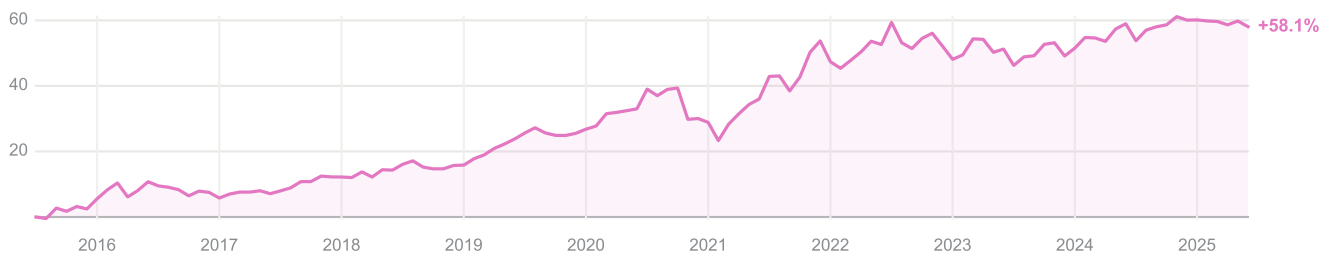
Historical Growth

CAGR +4.6%

171 ● **Cash Flow to Invested Capital**

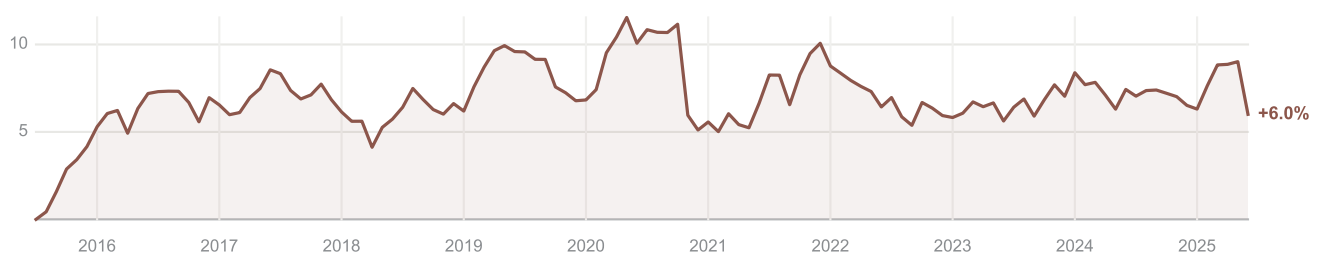
Earnings Quality

CAGR +4.6%

172 ● **Dividend Growth**

Historical Growth

CAGR +4.5%

173 ● **Return on Invested Capital**

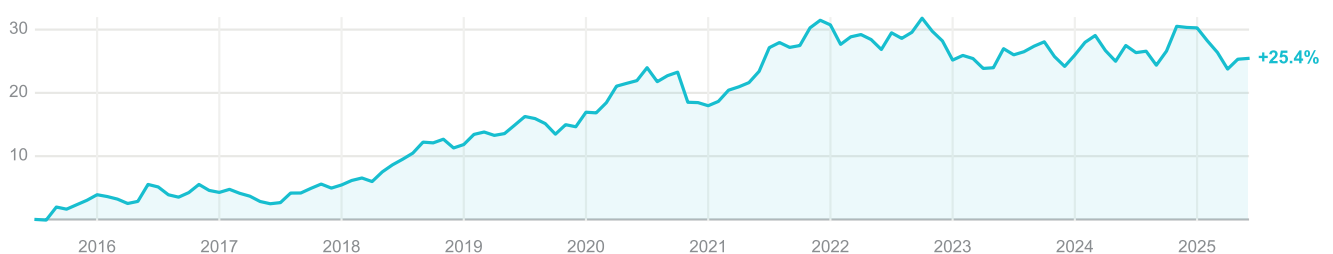
Capital Efficiency

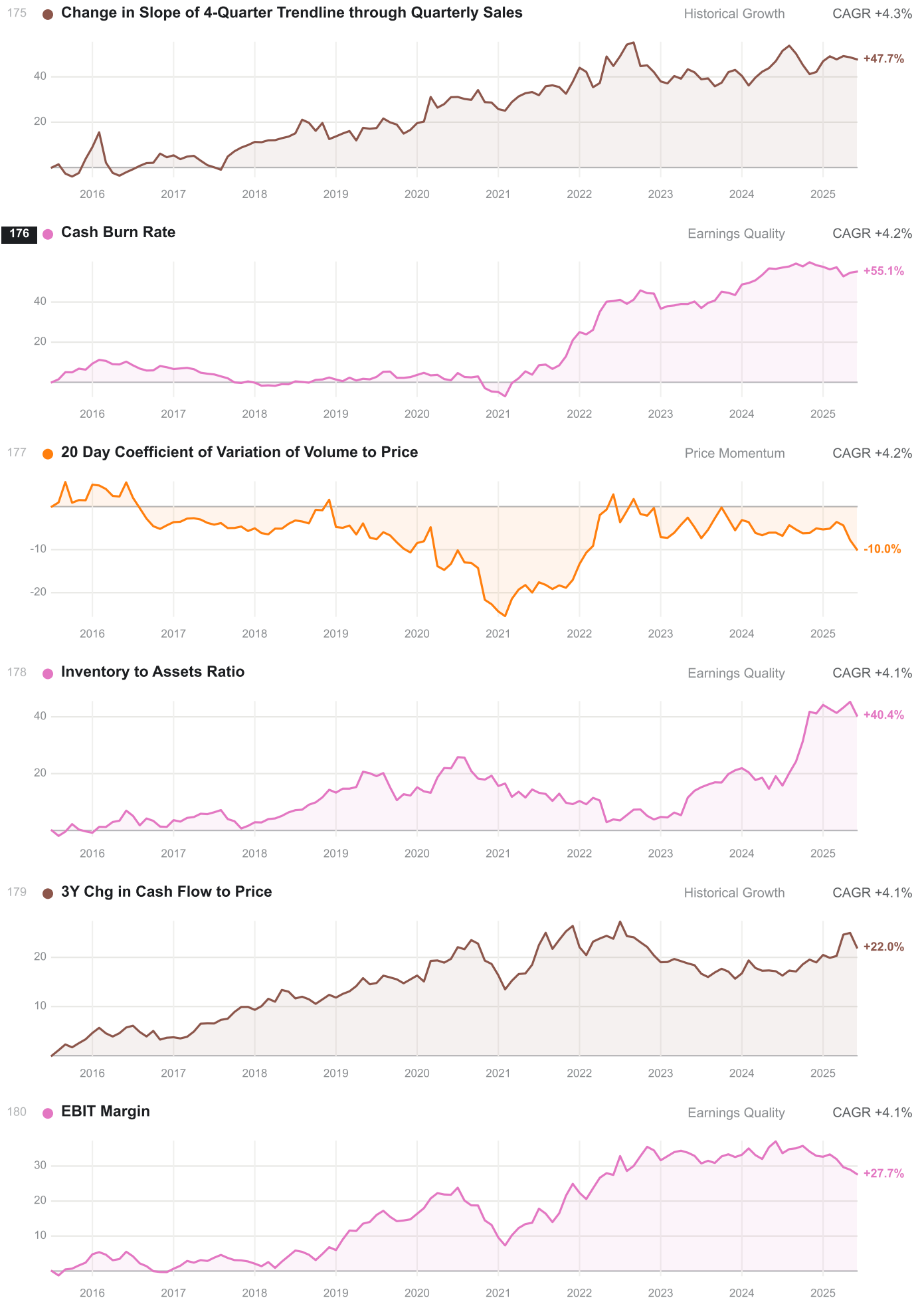
CAGR +4.5%

174 ● **Standardized Unexpected Earnings**

Analyst Expectations

CAGR +4.4%

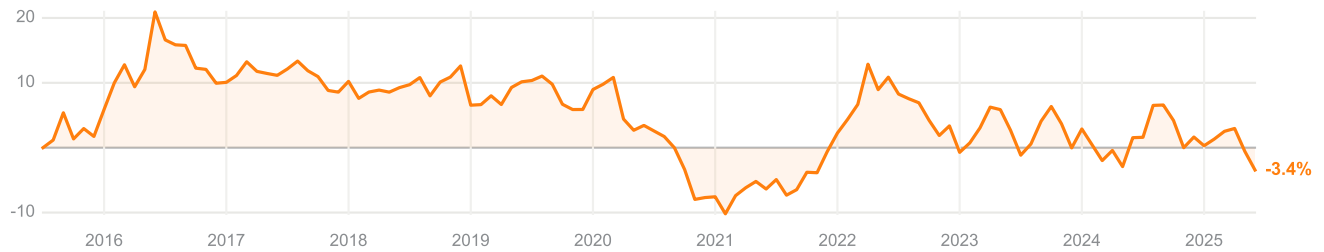




181 ● 60 Day Coefficient of Variation of Volume to Price

Price Momentum

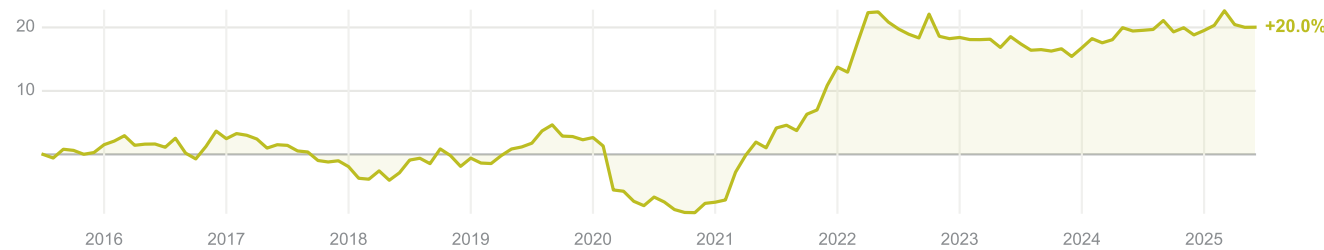
CAGR +4.0%



182 ● Degree of Financial Leverage

Capital Efficiency

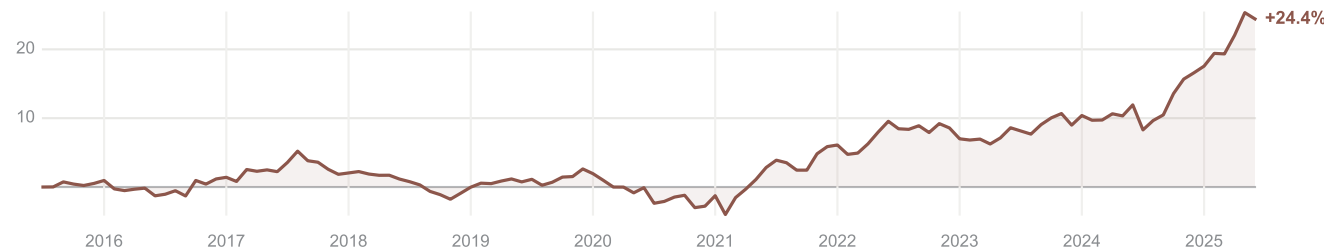
CAGR +4.0%



183 ● 3Y Chg in Sales Turnover

Historical Growth

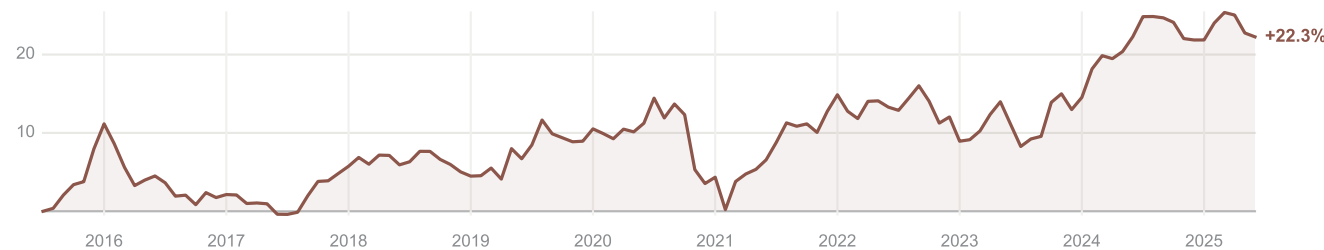
CAGR +4.0%



184 ● 3M Momentum in Trailing 12M Sales

Historical Growth

CAGR +4.0%



185 ● 3Y Average Annual Earnings Growth

Historical Growth

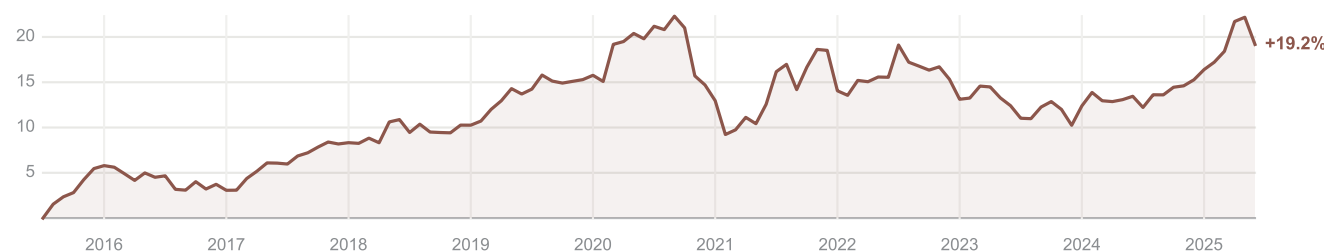
CAGR +4.0%

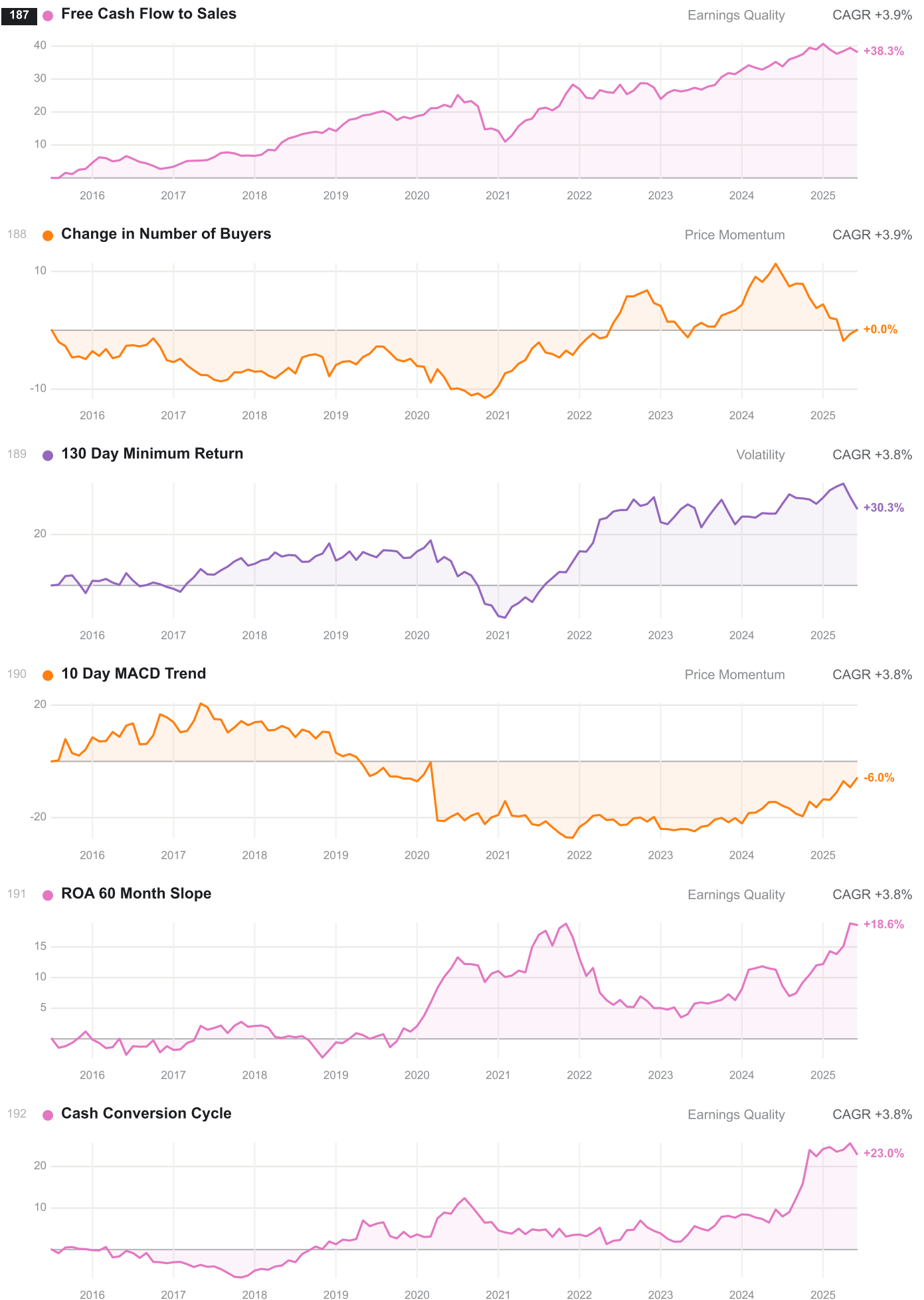


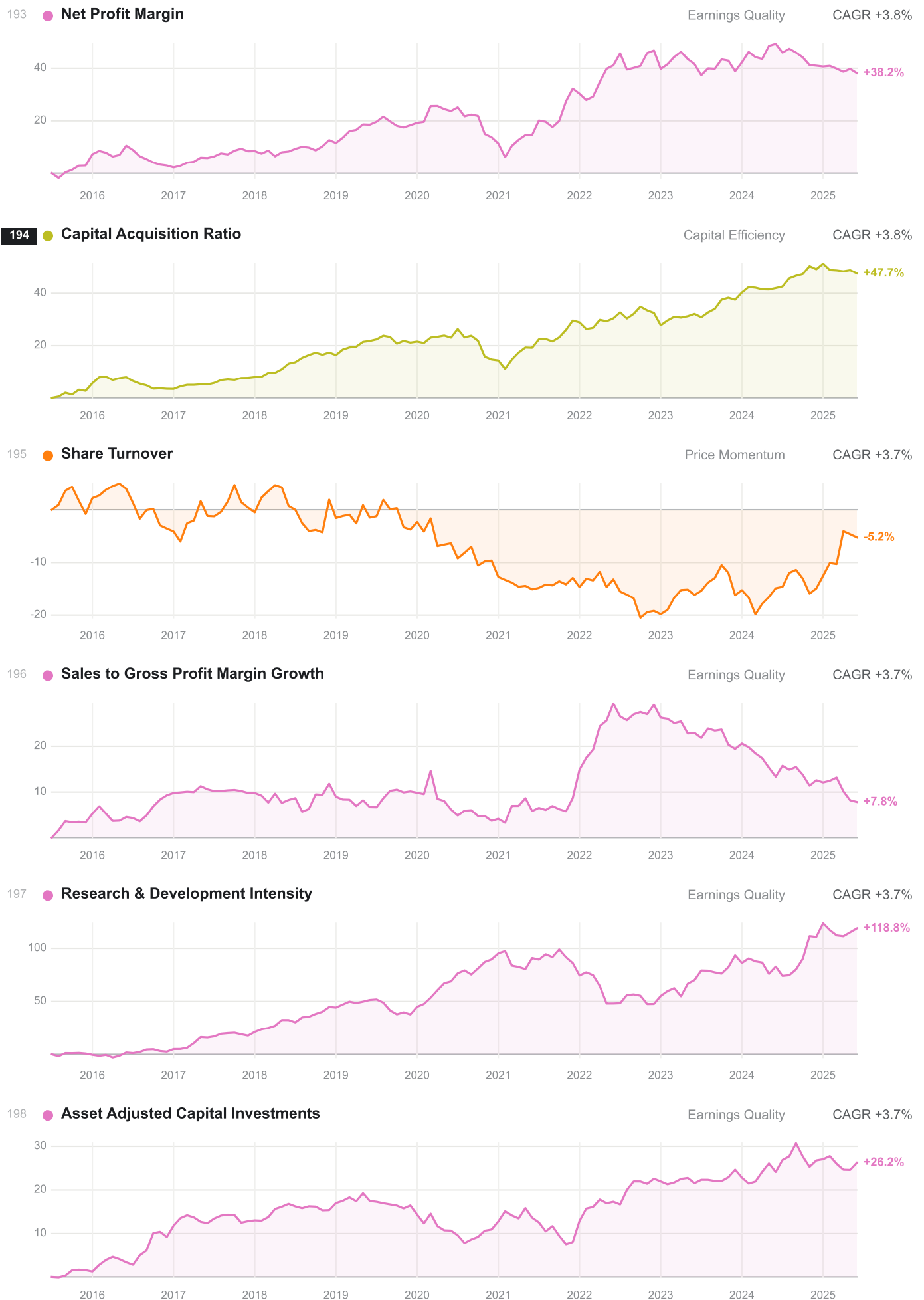
186 ● 3Y Chg in Earnings to Price

Historical Growth

CAGR +3.9%



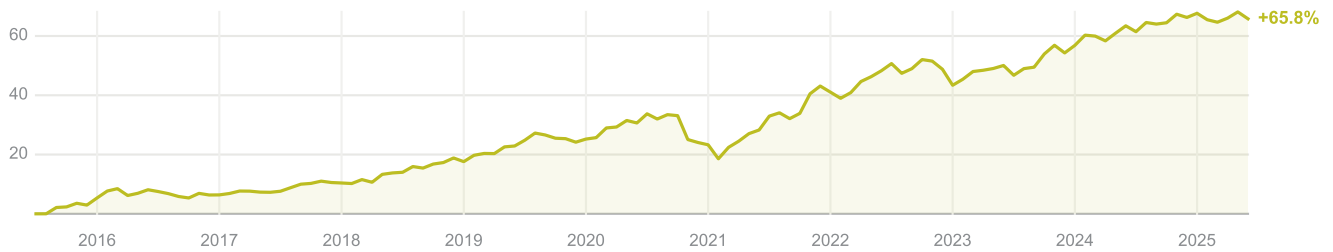




199 ● Cash Flow Return on Invested Capital

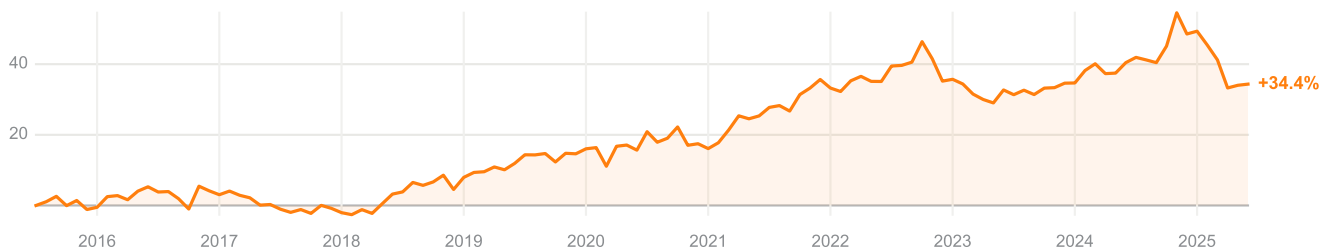
Capital Efficiency

CAGR +3.6%

**200 ● Institution Ownership Level**

Price Momentum

CAGR +3.6%

**201 ● Operating Cash Flow to Equity**

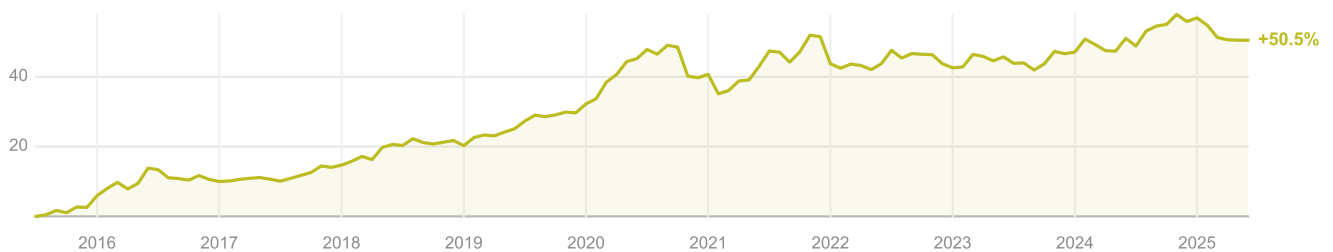
Capital Efficiency

CAGR +3.6%

**202 ● Gross Profit to Assets**

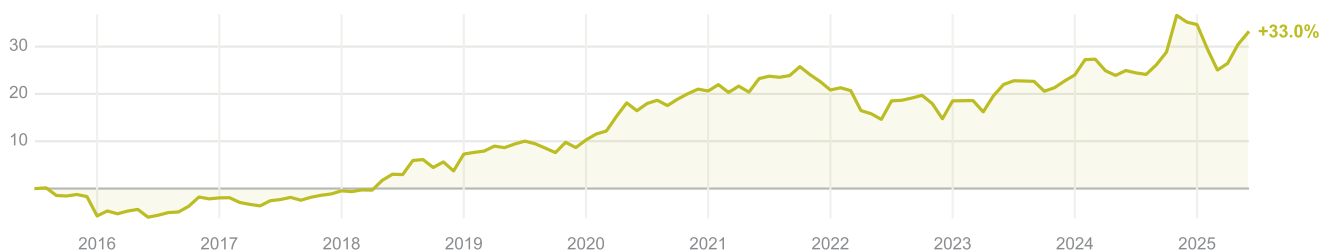
Capital Efficiency

CAGR +3.6%

**203 ● Payout Ratio**

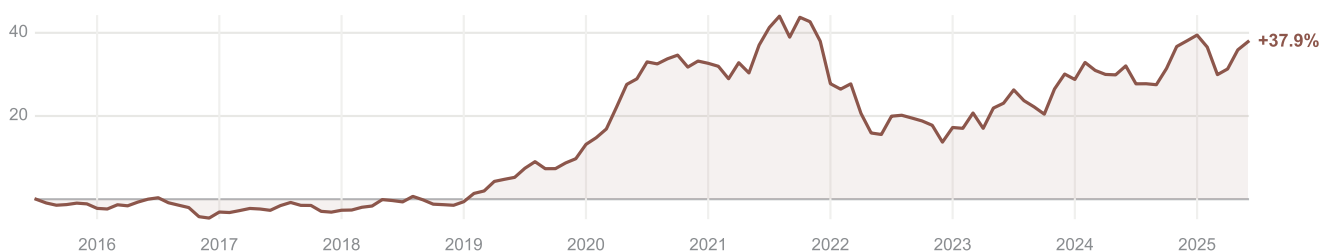
Capital Efficiency

CAGR +3.6%

**204 ● 3Y Average Annual Sales Growth**

Historical Growth

CAGR +3.5%



205 ● 3Y Chg in Asset Adjusted EPS

Historical Growth

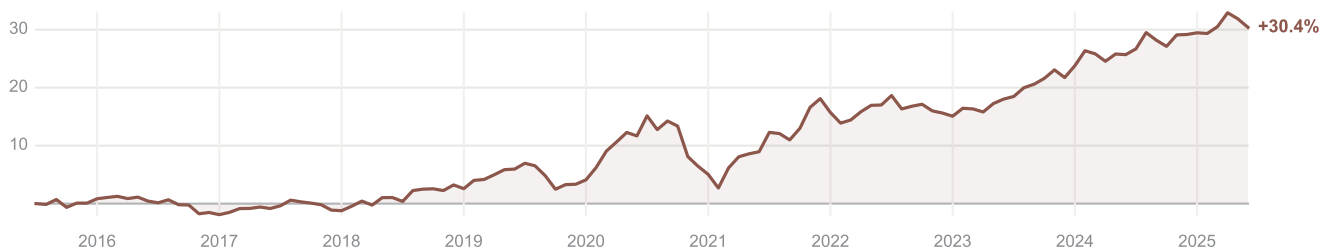
CAGR +3.5%



206 ● 3Y Chg in Free Cash Flow per Share

Historical Growth

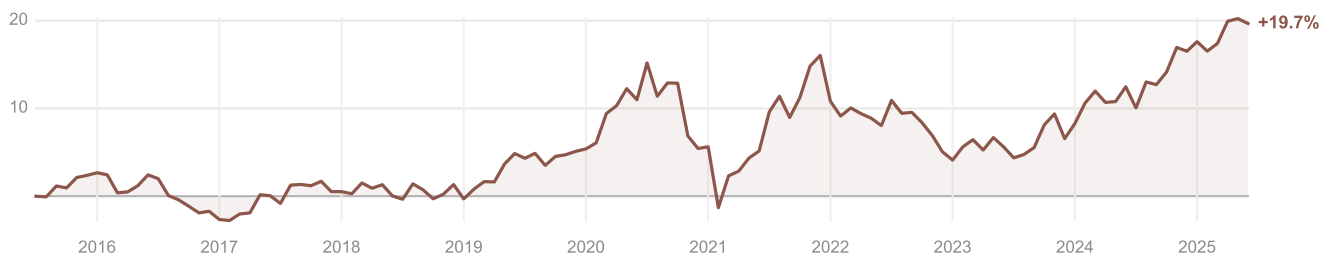
CAGR +3.5%



207 ● 1Y Chg in Earnings per Share

Historical Growth

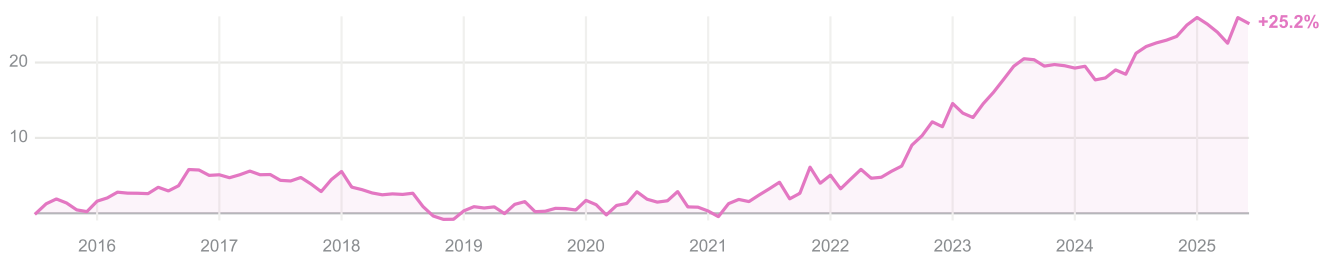
CAGR +3.5%



208 ● Unexpected Receivables Change

Earnings Quality

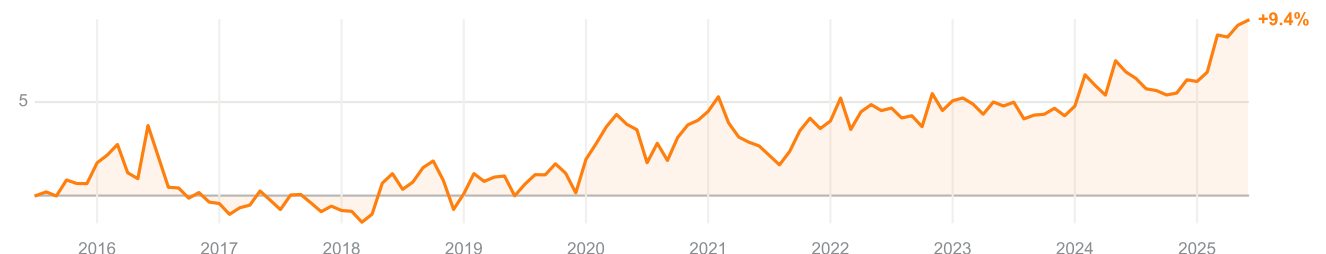
CAGR +3.5%



209 ● 6M Chg in 36M CAPM Alpha

Price Momentum

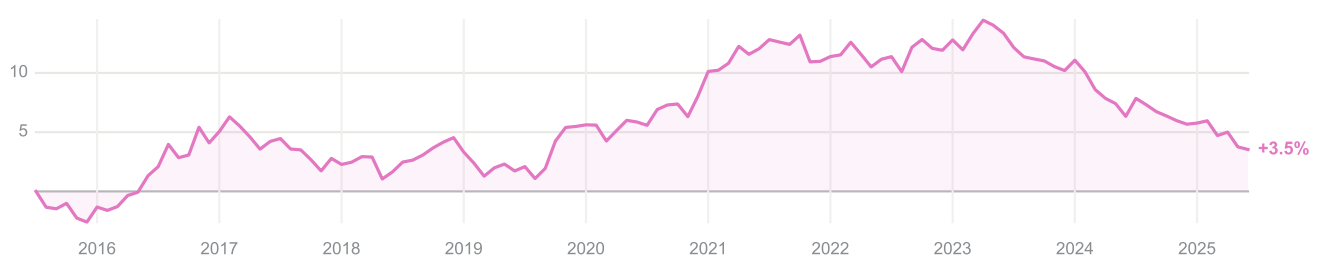
CAGR +3.5%



210 ● 1Y Chg in Sales to Earnings

Earnings Quality

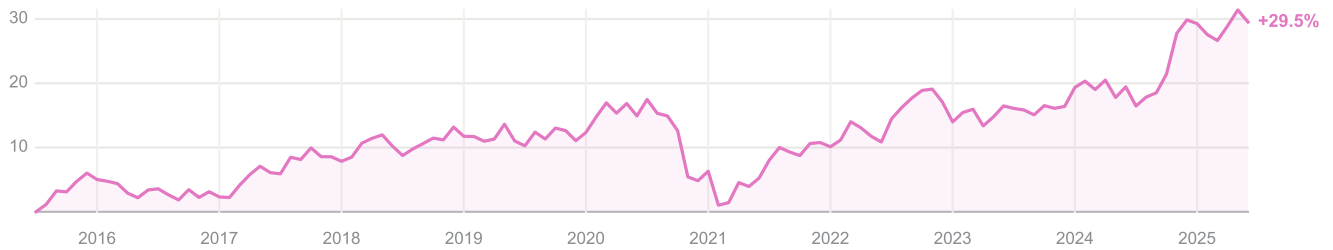
CAGR +3.5%



211 ● Operating Cash Flow Stability

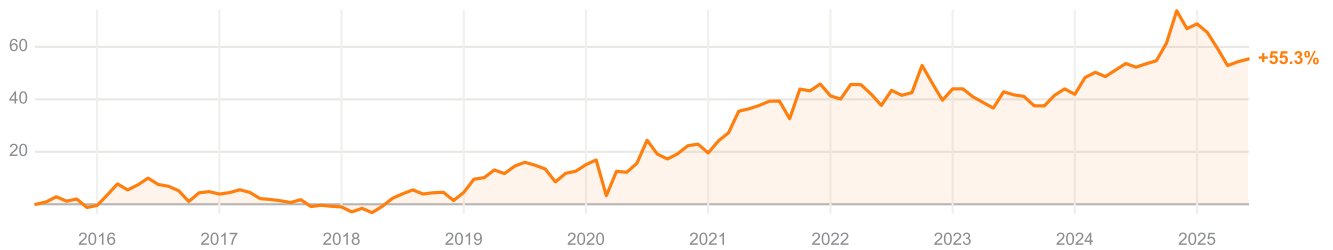
Earnings Quality

CAGR +3.4%

**212 ● 6 Month Average Share Turn Over**

Price Momentum

CAGR +3.4%

**213 ● EBIT to Price**

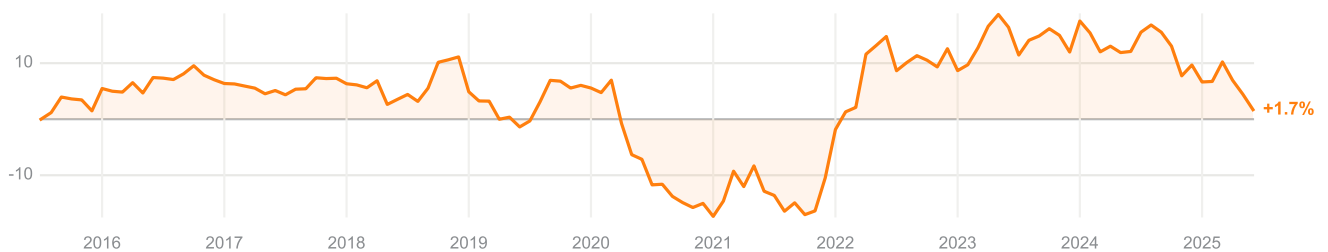
Valuation

CAGR +3.4%

**214 ● Ownership Breadth Stability of Institution Ownership by Holders**

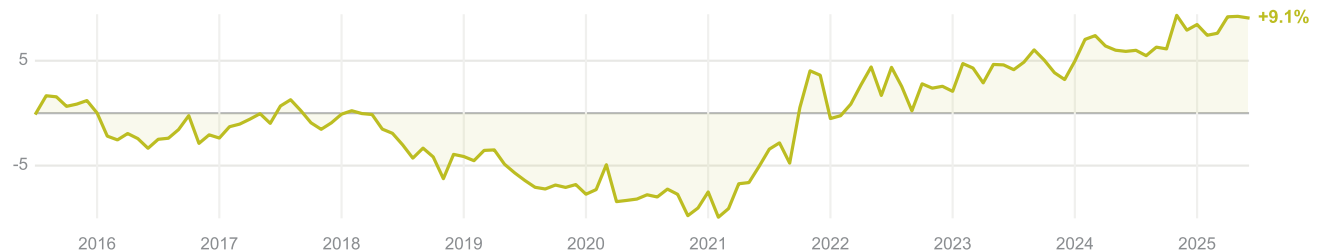
Price Momentum

CAGR +3.3%

**215 ● Year over Year Change of Total Debt**

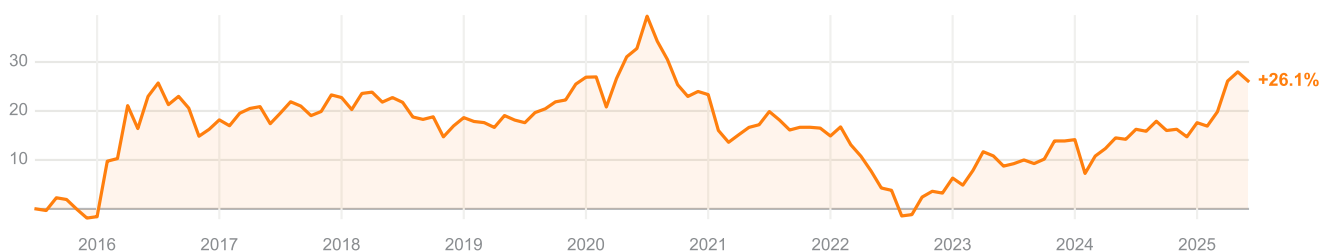
Capital Efficiency

CAGR +3.3%

**216 ● Foreign Institution Ownership**

Price Momentum

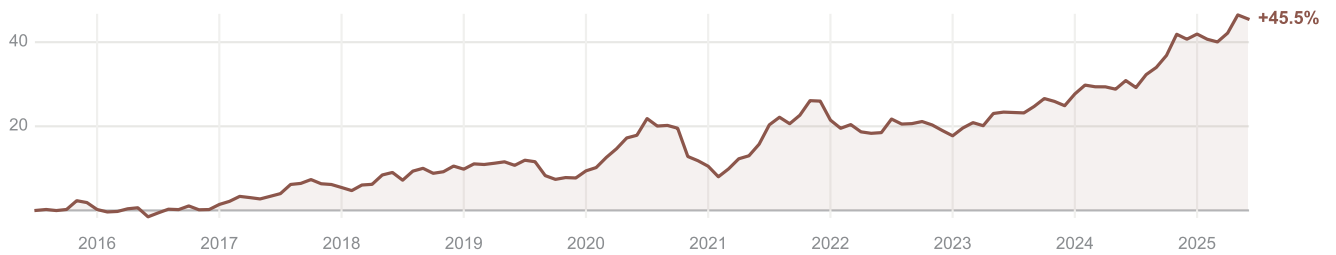
CAGR +3.3%



217 ● **3Y Chg in Asset Adjusted Free Cash Flow**

Historical Growth

CAGR +3.2%

218 ● **Free Cash Flow to Common Equity**

Valuation

CAGR +3.2%

219 ● **Sustainable Growth Rate**

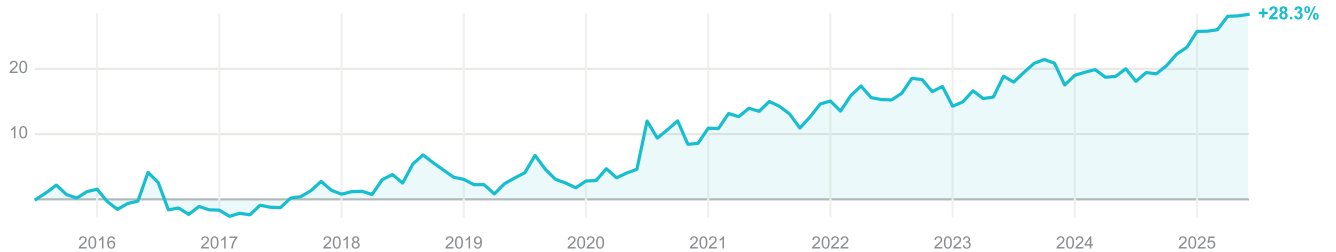
Historical Growth

CAGR +3.1%

220 ● **Adjusted Number of EPS FY1 Revisions**

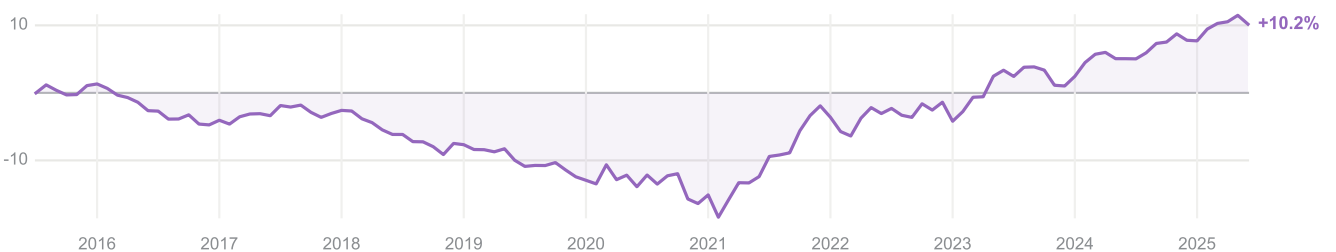
Analyst Expectations

CAGR +3.1%

221 ● **1Y Chg in Debt to Assets**

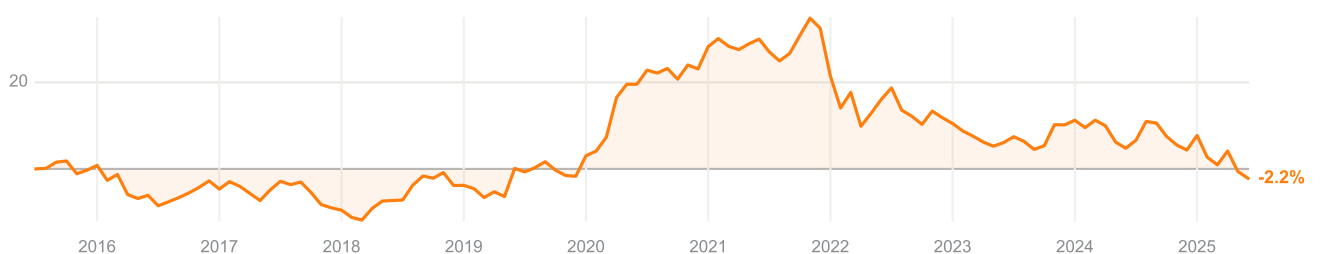
Volatility

CAGR +3.0%

222 ● **Standardized 36M Residual**

Price Momentum

CAGR +2.9%



223 ● 1Y Chg in Cash Flow per Share

Historical Growth

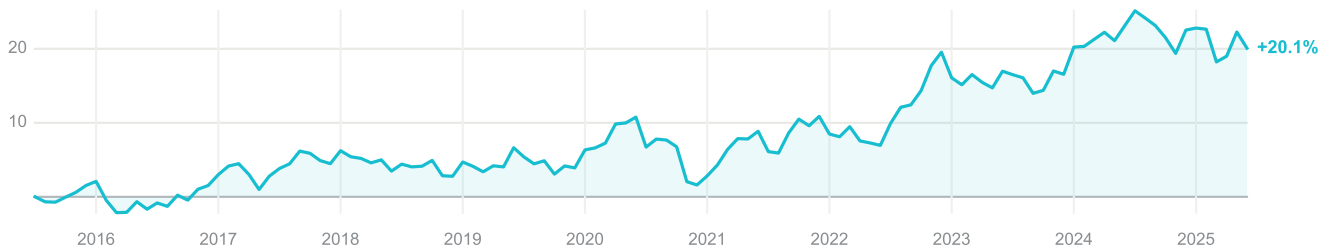
CAGR +2.8%



224 ● Target Price Gap to 6M EMA

Analyst Expectations

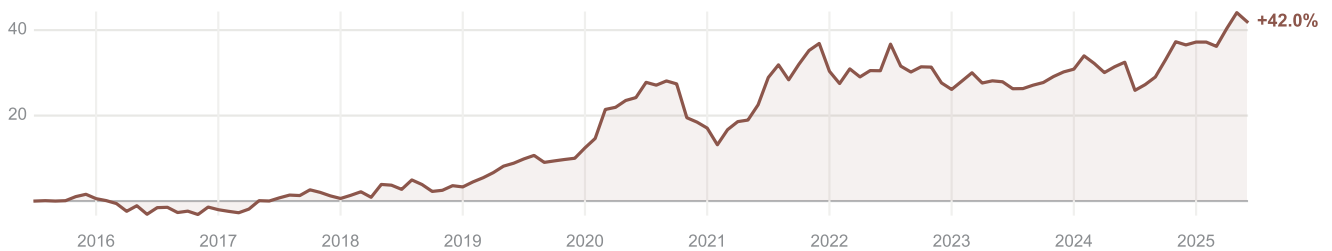
CAGR +2.8%



225 ● 3Y Chg in Asset Adjusted Cash Flow

Historical Growth

CAGR +2.8%



226 ● Net Current Assets to Price Ratio

Valuation

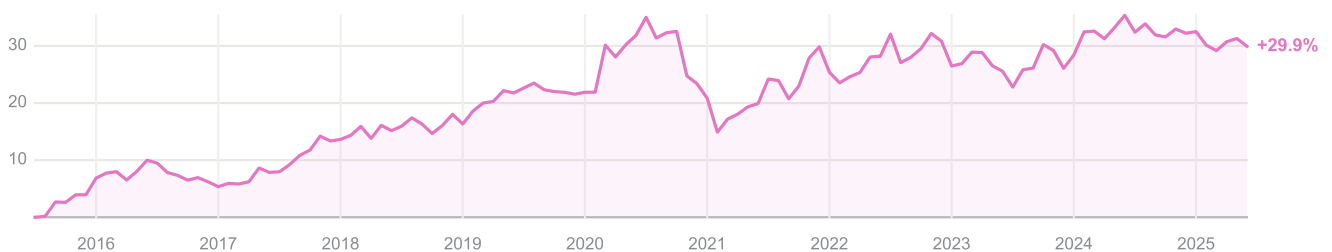
CAGR +2.7%



227 ● Pretax Income to Net Operating Assets

Earnings Quality

CAGR +2.7%



228 ● EBIT Estimate Dispersion

Analyst Expectations

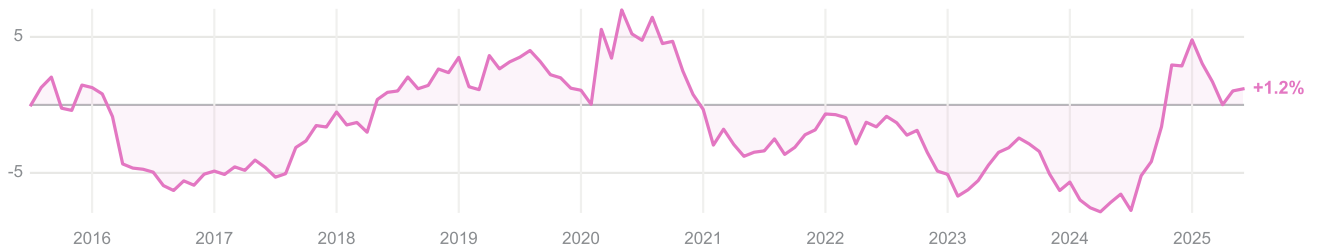
CAGR +2.7%



229 ● 1Y Chg in EPS to Operating Cash Flow

Earnings Quality

CAGR +2.6%



230 ● Working Capital Turnover Ratio

Capital Efficiency

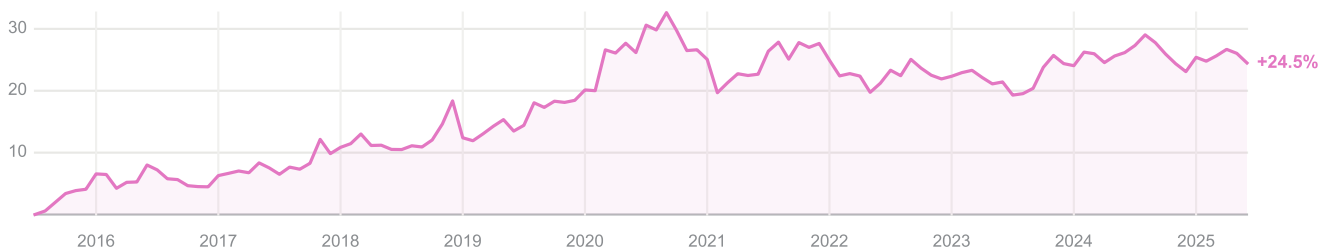
CAGR +2.6%



231 ● Change of Sales to Change of EPS

Earnings Quality

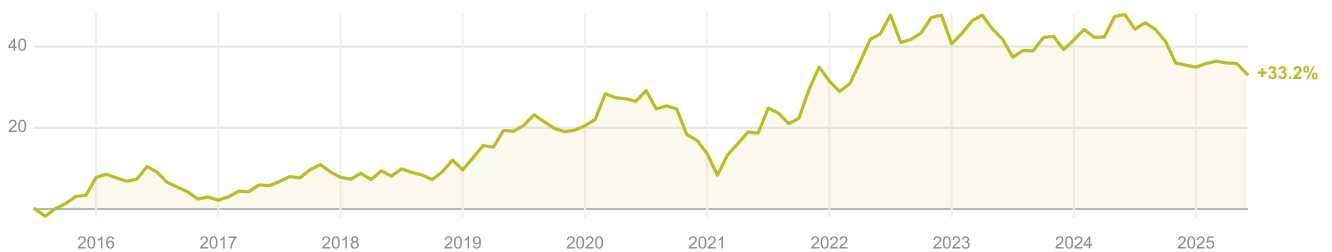
CAGR +2.5%



232 ● Pretax Profit Margin

Capital Efficiency

CAGR +2.5%



233 ● Sales Acceleration

Historical Growth

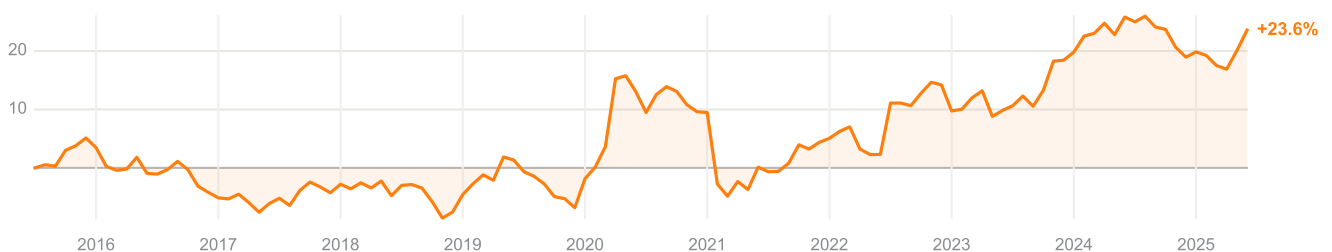
CAGR +2.4%

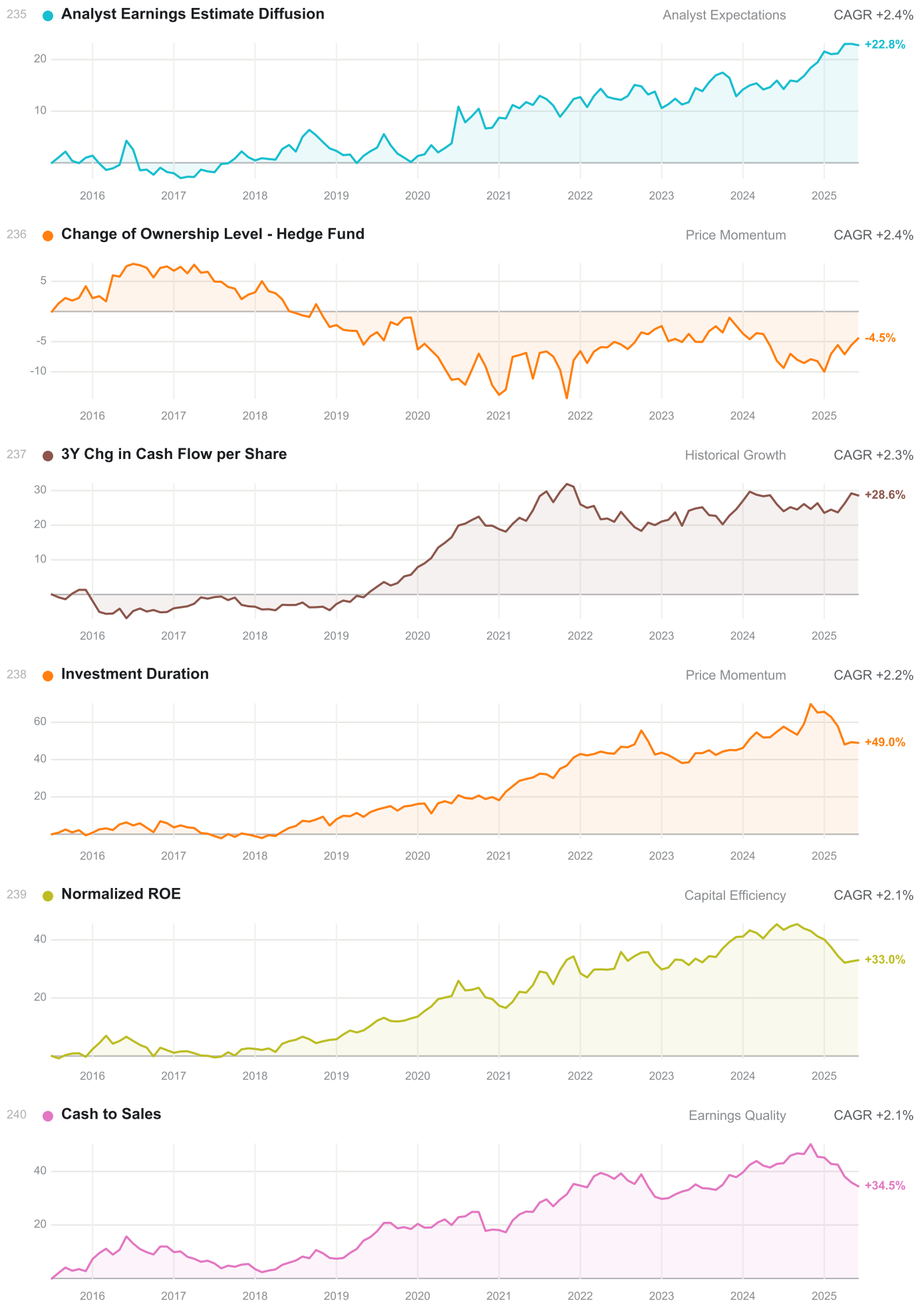


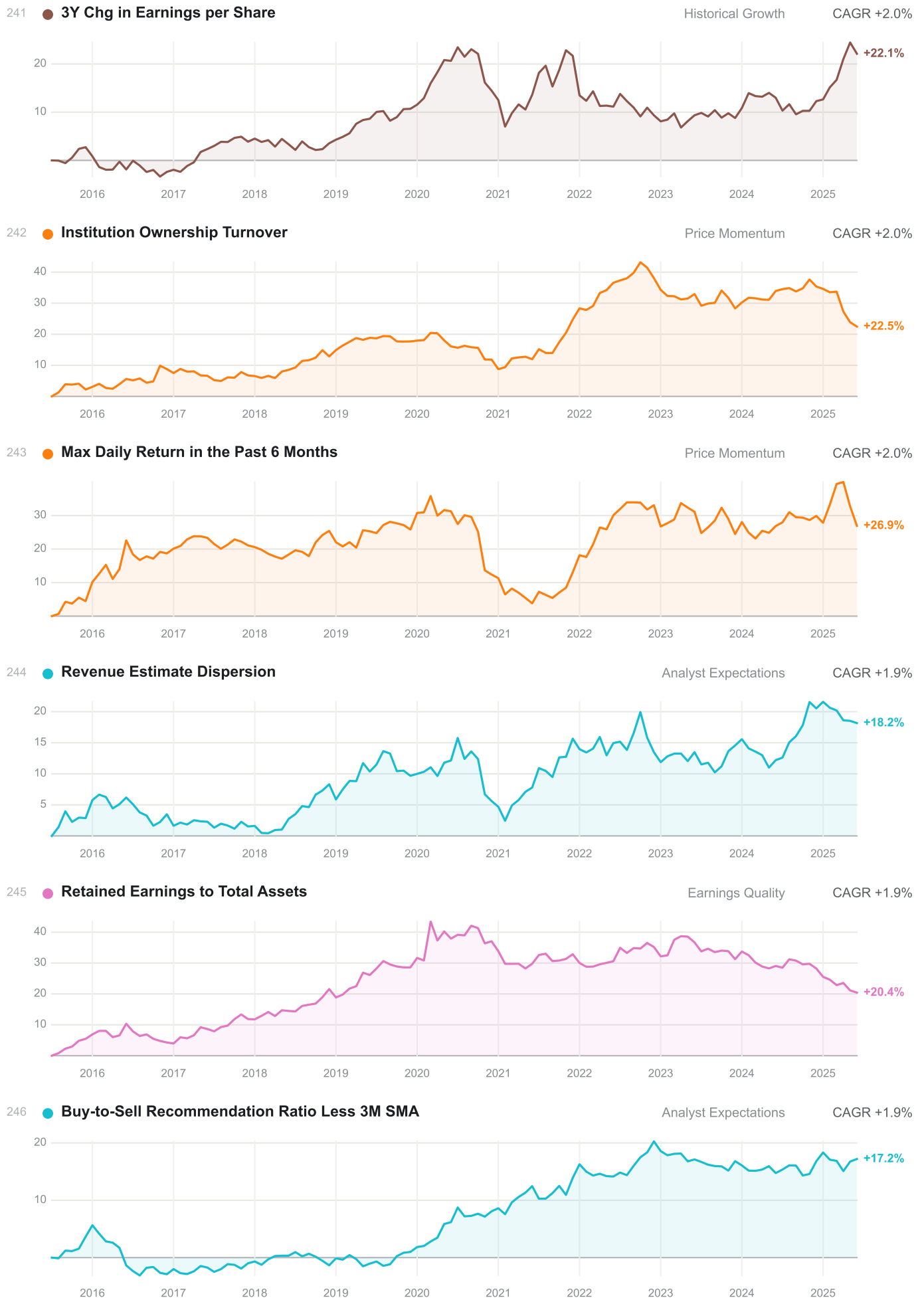
234 ● Risk Adjusted Relative Strength

Price Momentum

CAGR +2.4%



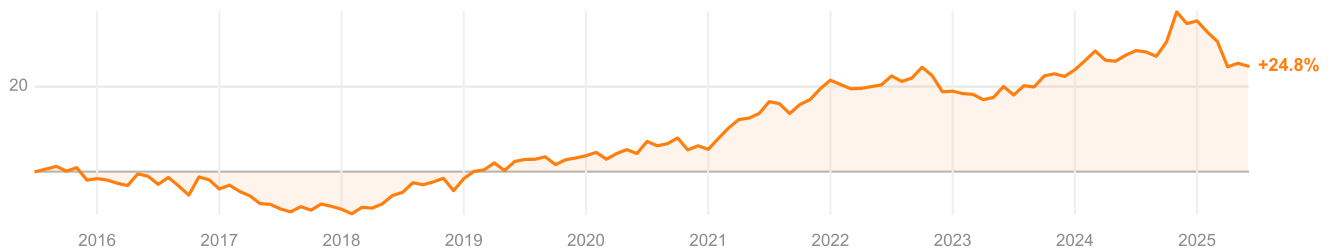




247 ● **Institution Ownership Concentration**

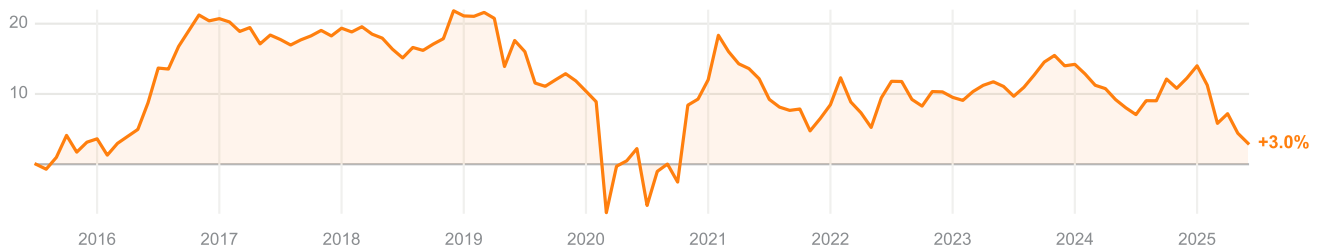
Price Momentum

CAGR +1.8%

248 ● **5 Day Price Reversal**

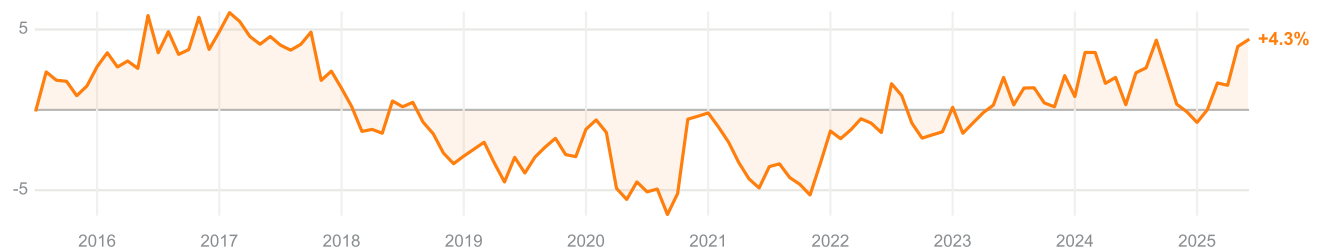
Price Momentum

CAGR +1.7%

249 ● **6M Chg in 18M CAPM Alpha**

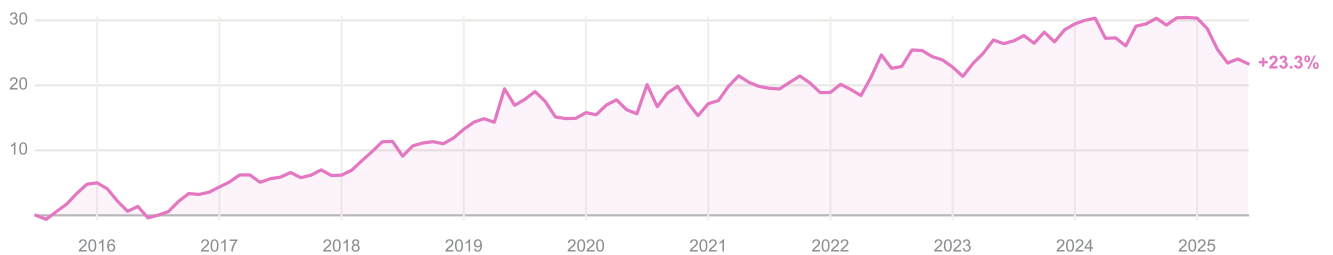
Price Momentum

CAGR +1.6%

250 ● **Year over Year Change of Inventory to Assets**

Earnings Quality

CAGR +1.6%

251 ● **6M Chg in 12M CAPM Alpha**

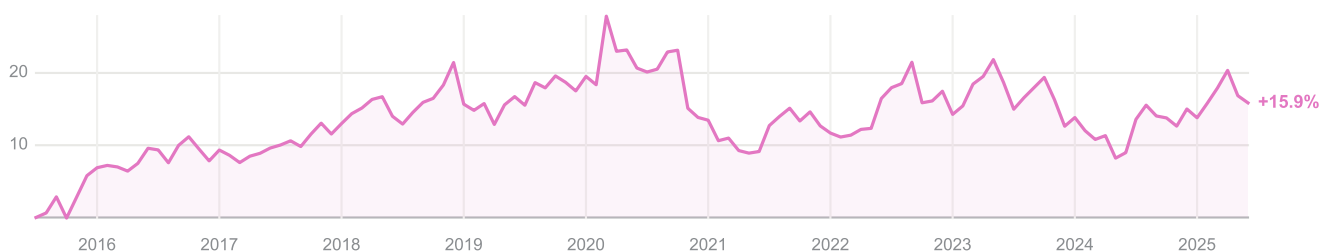
Price Momentum

CAGR +1.4%

252 ● **ROE 20 Qtr Standard Deviation**

Earnings Quality

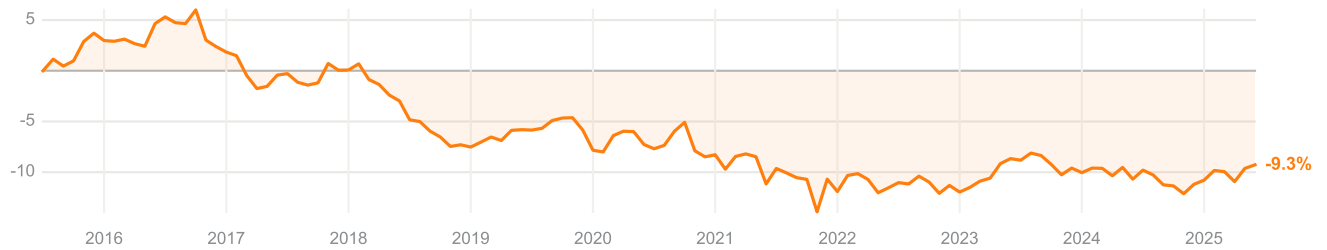
CAGR +1.4%



253 ● **Change of Ownership Level - All Managers**

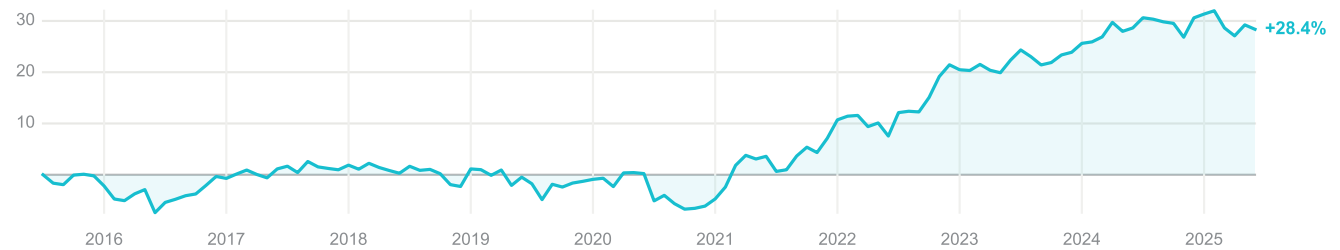
Price Momentum

CAGR +1.3%

254 ● **6M Chg in Target Price Gap**

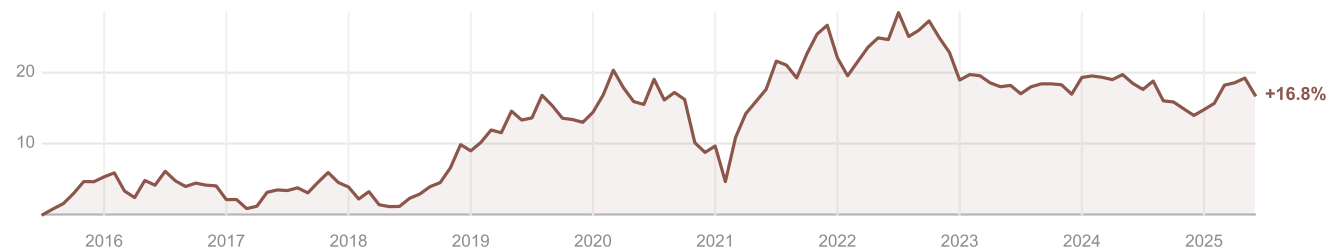
Analyst Expectations

CAGR +1.2%

255 ● **3Y Chg in Operating Margin**

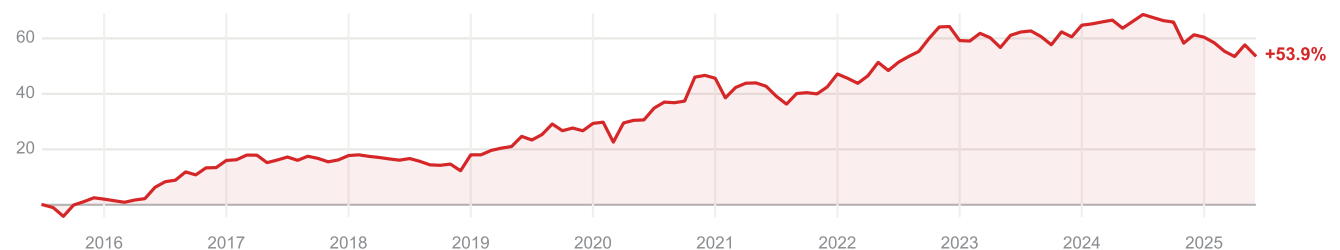
Historical Growth

CAGR +1.1%

256 ● **8-Week Change in 12-Month Forward Earnings Consensus Estimate/Price**

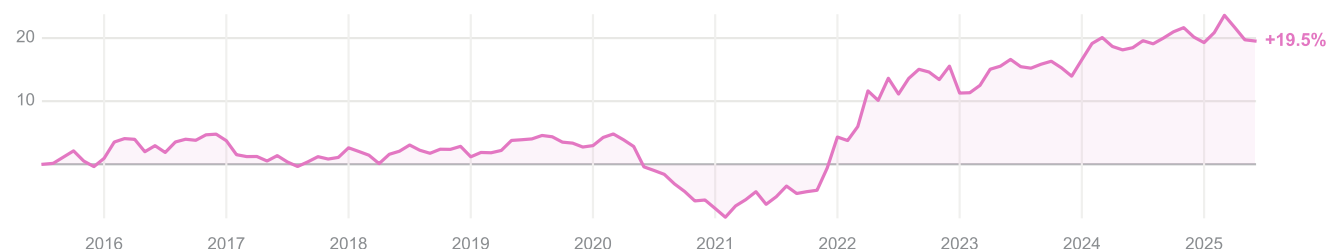
Valuation

CAGR +1.0%

257 ● **Change in TTM Depr. to Capex**

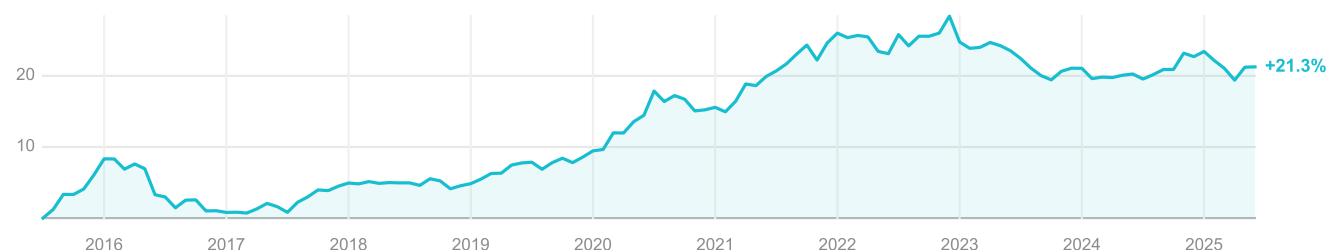
Earnings Quality

CAGR +0.9%

258 ● **6M Avg Chg 1M Recommendation**

Analyst Expectations

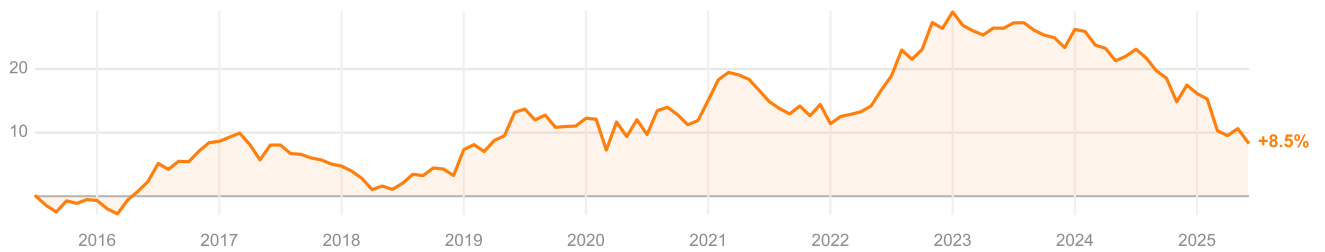
CAGR +0.7%



259 ● **14-Day Relative Strength Index**

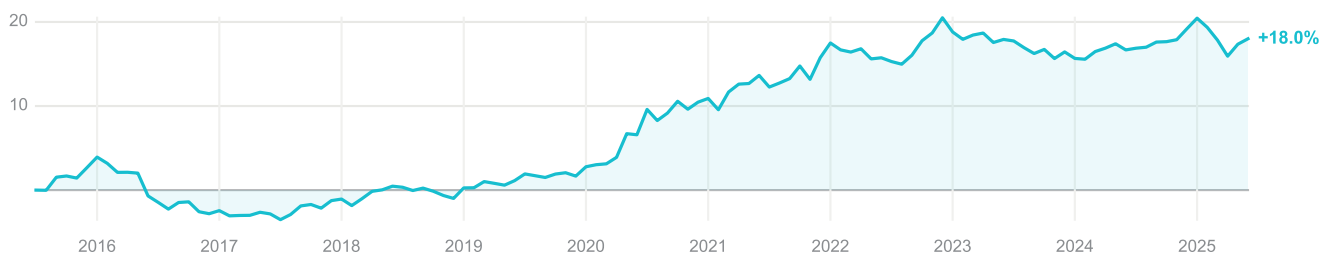
Price Momentum

CAGR +0.5%

260 ● **Buy-to-Sell Recommendation Ratio Less 3M EMA**

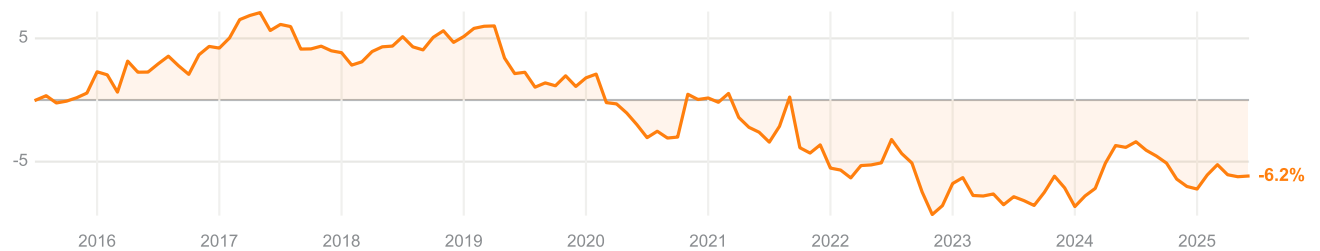
Analyst Expectations

CAGR +0.3%

261 ● **1Y Chg in Share Turnover**

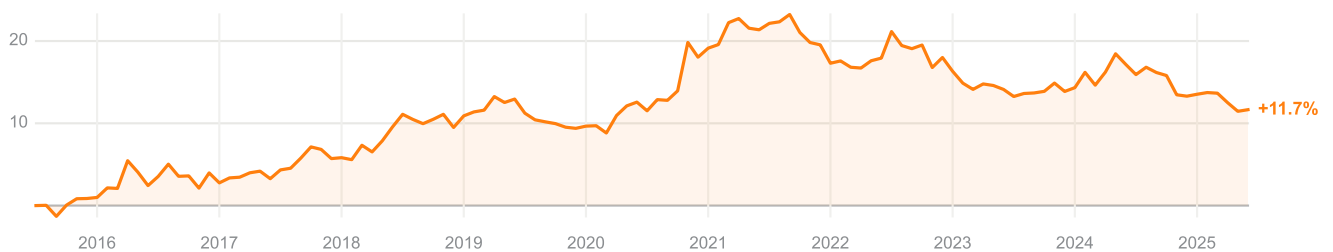
Price Momentum

CAGR +0.2%

262 ● **50 Day Volume Signal**

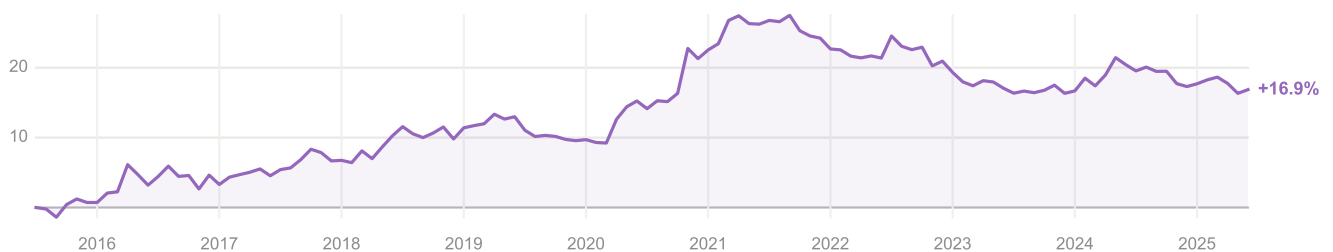
Price Momentum

CAGR -0.6%

263 ● **Adj 50 Day Volume Signal**

Volatility

CAGR -0.6%

264 ● **4-Week Change in 12-Month Forward Earnings Consensus Estimate/Price**

Valuation

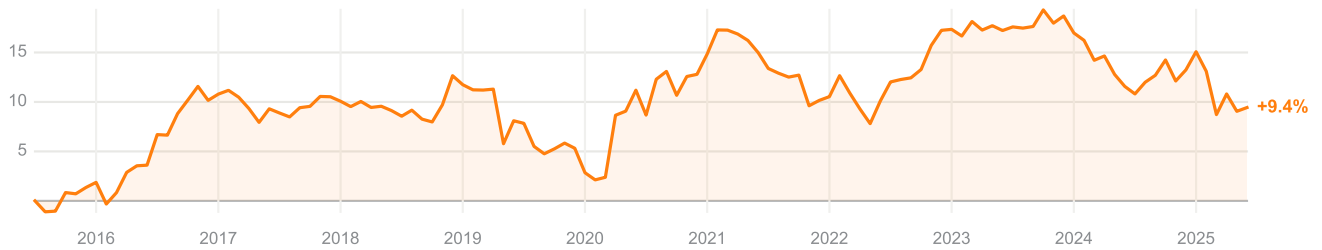
CAGR -0.7%



265 ● 5-Day Money Flow/Volume

Price Momentum

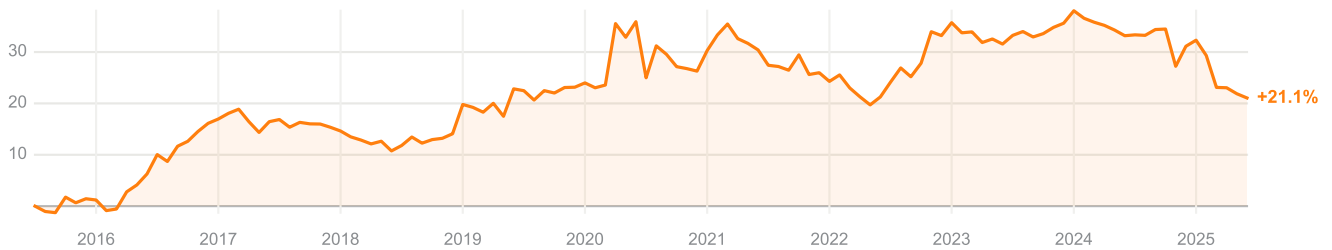
CAGR -1.1%



266 ● 20-Day Lane's Stochastic Indicator

Price Momentum

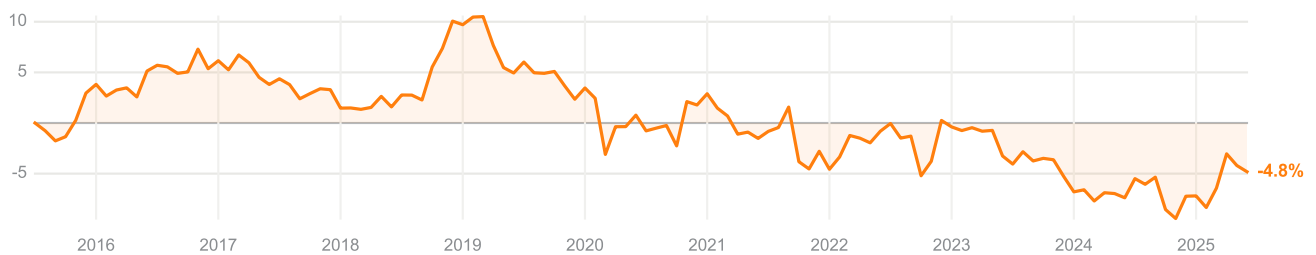
CAGR -1.4%



267 ● 5 Day Volume Signal

Price Momentum

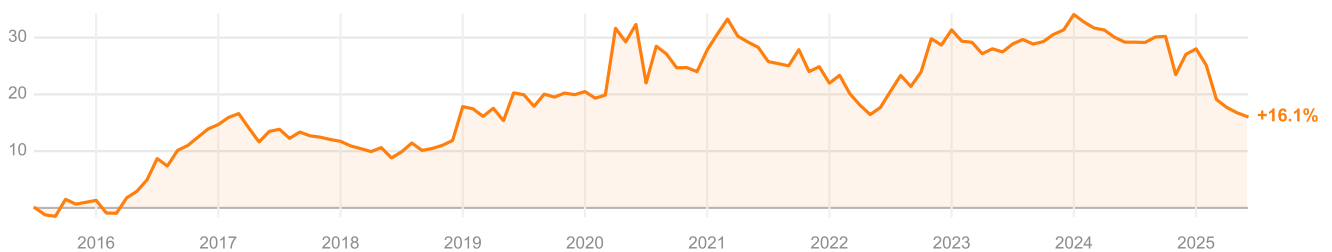
CAGR -1.9%



268 ● 1M Price High - 1M Price Low

Price Momentum

CAGR -1.9%



269 ● 1M Price Reversal

Price Momentum

CAGR -4.0%

